WSQ · SOFTWARE DESIGN

Build ASP.NET Web Apps with Vibe Coding



Course Code: TGS-2023040477

Software Design · ICT-DES-3005-1.1

Trainer: Dr. Alfred Ang

Version 6.1 · 12 September 2026

Tertiary Infotech Academy Pte Ltd
UEN 201200696W

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Digital Attendance (Mandatory)

1

AM attendance

Scan the SSG QR code shown by the trainer.

2

PM attendance

Complete the second attendance event after lunch.

3

Assessment attendance

Complete the dedicated assessment attendance.

4

Evidence

Minimum 75% attendance and competent assessment are funding conditions.

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About the Trainer



General Trainer template

Complete before delivery

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

AREAS OF EXPERTISE _____

TRAINING & INDUSTRY EXPERIENCE _____

CONTACT _____

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About the Trainer

AA

Dr. Alfred Ang

Principal Trainer · software and AI-enabled development

NAME	Dr. Alfred Ang
TITLE / DESIGNATION	Principal Trainer
QUALIFICATIONS	Doctorate · industry certifications
AREAS OF EXPERTISE	ASP.NET · software design · AI coding
TRAINING & INDUSTRY EXPERIENCE	Adult education and industry practice
CONTACT	enquiry@tertiaryinfotech.com

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LEARNER INTRODUCTION

Let's Know Each Other

1

Name + role

Share your current work context.

2

Web experience

Name one app or API you have used.

3

ASP.NET baseline

Share your .NET/C# exposure.

4

Learning goal

Name one CRM outcome you want to build.

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LEARNING ENVIRONMENT

Ground Rules

1 **Participate**

Ask, explain and test actively.

2 **Respect**

One conversation; challenge ideas, not people.

3 **Be punctual**

Return from breaks on time.

4 **Protect data**

Use only provided mock CRM records.

5 **Verify AI**

Inspect every generated diff and dependency.

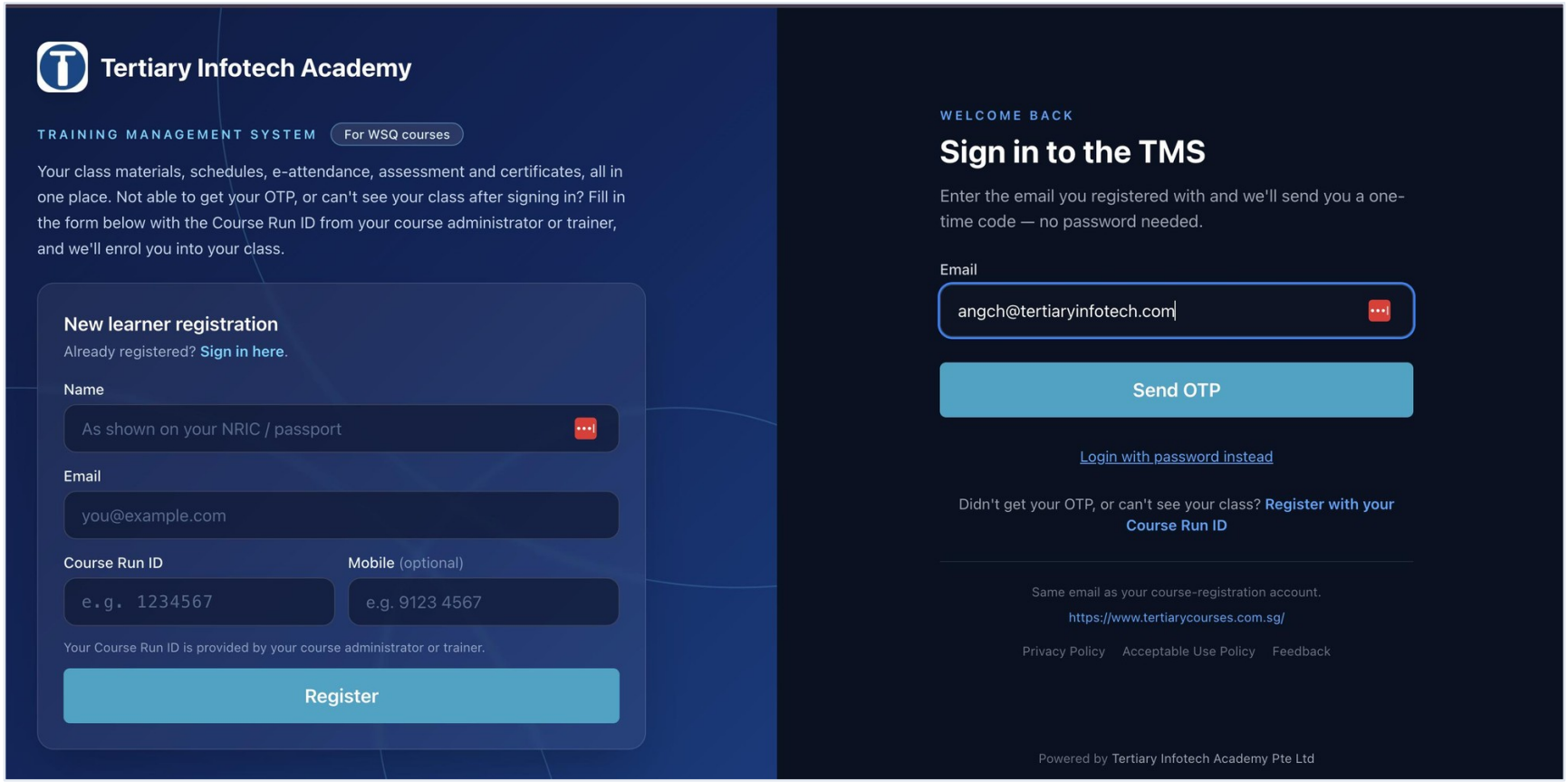
6 **Attend**

Complete AM, PM and assessment attendance.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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COURSE ADMINISTRATION · 09:00–18:00

Lesson Plan · Day 1 · Part 1/2

09:00–09:30	Administration, attendance, introductions and assessment briefing	Facilitation
09:30–10:45	Topic 1 — ASP.NET Core runtime and component design	Lecture, demonstration, questioning
10:45–10:55	Morning break	Break
10:55–12:30	Activities 1–2 — requirements, scaffold and controlled vibe change	Practical
12:30–13:30	Lunch	Break
13:30–14:50	Topic 2 — accessible UI and organisational practice	Lecture, demonstration, questioning
14:50–15:40	Activities 3–4 — accessible shell and audit	Practical

Continued on the next schedule slide.

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COURSE ADMINISTRATION · 09:00–18:00

Lesson Plan · Day 1 · Part 2/2

15:40–15:50	Afternoon break	Break
15:50–17:00	Topic 3 — Razor Pages controls, binding and persistence	Lecture and demonstration
17:00–17:50	Activity 5 — create and list with Razor Pages	Practical
17:50–18:00	Day 1 recap and PM attendance	Reflection

Tea breaks sit within facilitated time; lunch is excluded. Final assessment runs Day 2, 16:00–18:00.

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COURSE ADMINISTRATION · 09:00–18:00

Lesson Plan · Day 2 · Part 1/2

09:00–09:15	Day 1 recap and AM attendance	Review
09:15–10:05	Topic 3 — validation, Tag Helpers and antiforgery	Lecture and demonstration
10:05–10:55	Activity 6 — validation and antiforgery	Practical
10:55–11:05	Morning break	Break
11:05–12:10	Topic 4 — controller APIs and HTTP contracts	Lecture, demonstration, questioning
12:10–13:00	Activity 7 — controller API	Practical
13:00–14:00	Lunch	Break

Continued on the next schedule slide.

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COURSE ADMINISTRATION · 09:00–18:00

Lesson Plan · Day 2 · Part 2/2

14:00–14:55	Activity 8 — SQLite, OpenAPI and integration tests	Practical
14:55–15:35	Topic 5 — hosting, Azure and design evidence	Lecture and demonstration
15:35–15:45	Afternoon break	Break
15:45–16:00	Activities 9–10 briefing and course synthesis	Demonstration and reflection
16:00–17:00	**Written Assessment — Short-Answer Questions (open book)**	Summative assessment
17:00–18:00	**Practical Performance Assessment (open book)**	Summative assessment

Tea breaks sit within facilitated time; lunch is excluded. Final assessment runs Day 2, 16:00–18:00.

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Learning Outcomes

1

LO1

Determine basic software components using ASP.NET framework to meet functional specifications.

2

LO2

Utilize ASP.NET framework to develop web apps in adherence to organizational practices.

3

LO3

Select essential ASP.NET controls and features to meet software design requirements.

4

LO4

Examine the interoperability and functionality of ASP.NET software components.

5

LO5

Generate ASP.NET design documentation aligned with user specifications.

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Skills Framework Alignment

1

K1 / A1

Requirements and component boundaries · Activities 1–2

2

K2 / A2

Design tools, accessibility and controlled practice · Activities 3–4

3

K3 / A3

Razor controls, binding, validation and antiforgery · Activities 5–6

4

K4 / A4

Controller API, status/schema interoperability · Activities 7–8

5

K5 / A5

Deployment and traceable design documentation · Activities 9–10

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BEFORE THE ASSESSMENT

Briefing for Assessment

1 **Open book**

Slides, Learner Guide and approved Activity material only.

2 **Individual evidence**

No discussion, messaging, screenshots or recording.

3 **WA first**

Five short-answer questions map K1–K5; 60 minutes.

4 **PP second**

Five practical tasks map A1–A5; 60 minutes.

5 **Submission**

Upload the required evidence on the LMS.

6 **Clarification**

Assessor may use oral clarification only to close a minor gap.

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SUMMATIVE · INDIVIDUAL · OPEN BOOK

Assessment and Funding

1 Written Assessment

WA-SAQ · 1 hour · K1–K5

2 Practical Performance

PP · 1 hour · A1–A5

3 Competency

Complete both modes and be assessed Competent.

4 Funding condition

At least 75% verified SSG attendance; eligibility and terms apply.

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MANDATORY SEQUENCE

Assessment Flow



Courseware and assessment: https://lms-tms.tertiaryinfotech.com/

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ONE CUMULATIVE LOCAL CRM · 10 SNAPSHOTS

Access the Hands-On Activities

1

Option A · clone

git clone https://github.com/tertiarycourses/TGS-2023040477-Build-ASP.NET-Web-Apps-with-Vibe-Coding.git

2

Option B · ZIP

GitHub · Code · Download ZIP

3

Run locally

Each Activity has src/LocalCRM plus its own mock-data folder.

4

Start point

Open activities/README.md and choose Activity 01.

https://github.com/tertiarycourses/TGS-2023040477-Build-ASP.NET-Web-Apps-with-Vibe-Coding

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Modern ASP.NET Core · Legacy Boundary

1 Current course stack

ASP.NET Core 10: Razor Pages, controllers, OpenAPI, EF Core and Kestrel.

2 ASP.NET 4.x

Web Forms/MVC 5 depend on .NET Framework/System.Web; historical comparison only.

3 Classic ASP

.asp and VBScript are a separate legacy technology; not taught as current Core.

4 Source rule

Microsoft Learn current docs outrank secondary legacy tutorials.

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TOPIC 1 · LO1 · K1 / A1

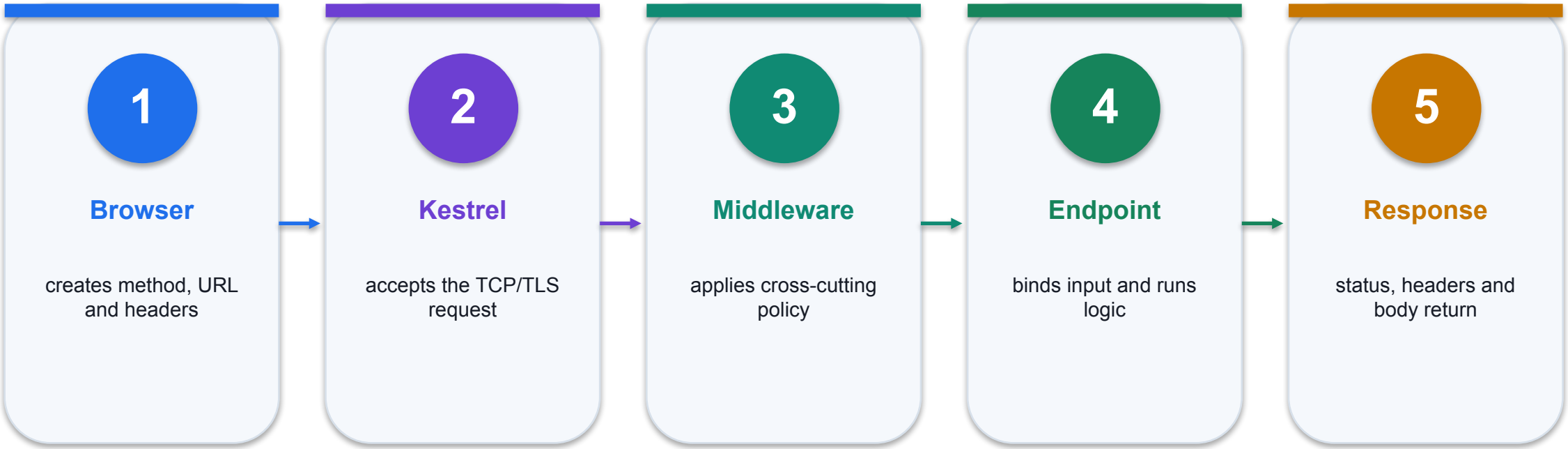
01

Design Components and the ASP.NET Core Runtime

requirements → component boundaries → request pipeline → verified scaffold

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HTTP request and response contract



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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HTTP request and response contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  app.MapGet("/health", () => Results.Ok(new { status =
   "ok" }));
```

1 **Method**

GET reads; POST creates; PUT replaces; DELETE removes

2 **Target**

/contacts/42 plus query string

3 **Headers**

Accept, Content-Type, Authorization, trace context

4 **Body**

JSON or form payload; never implied by a GET

K1 / A1

Full procedures: Learner Guide

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TOPIC 1 · DESIGN DETAIL

HTTP request and response contract · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Method	GET reads; POST creates; PUT replaces; DELETE removes	Where is this enforced?
Target	/contacts/42 plus query string	What is observable?
Headers	Accept, Content-Type, Authorization, trace context	Who owns the value?
Body	JSON or form payload; never implied by a GET	What rejects invalid input?
Status	2xx success, 4xx caller issue, 5xx server failure	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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HTTP request and response contract · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

404	route template did not match
405	path matched but method did not
415	Content-Type cannot be read
500	unhandled endpoint exception

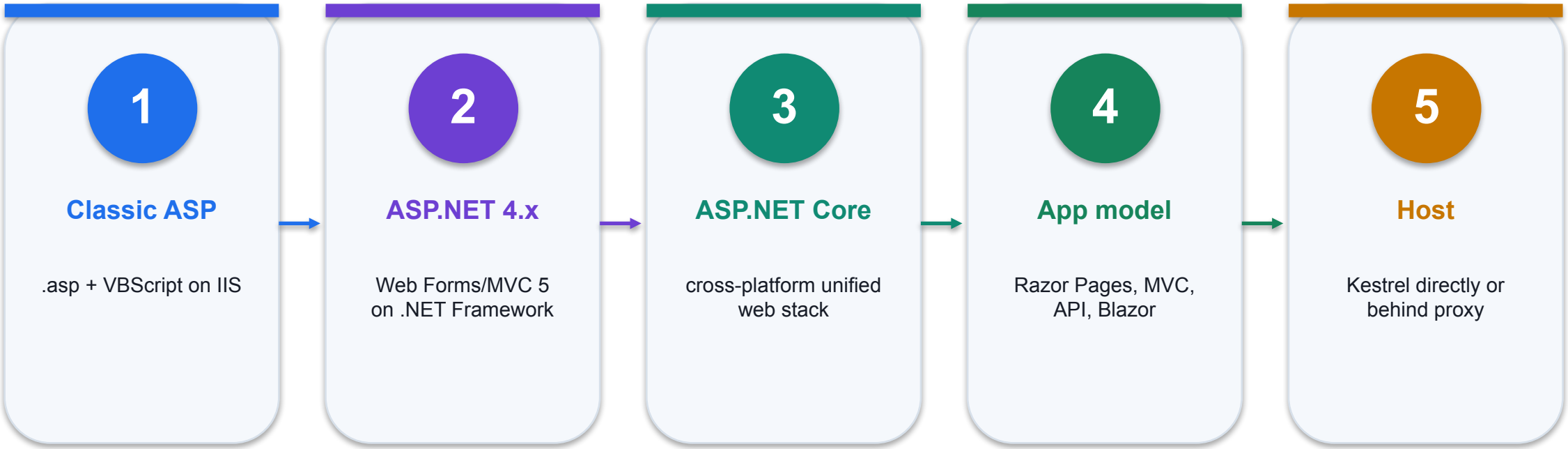
ACCEPTANCE EVIDENCE

- ✓ curl -i /health returns HTTP 200
- ✓ Content-Type is application/json
- ✓ The response body matches the declared contract

Record URL · command · status · build SHA

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ASP.NET technology family boundary



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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ASP.NET technology family boundary · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  <TargetFramework>net10.0</TargetFramework>
2  <Nullable>enable</Nullable>
```

1 **Classic ASP**
Server-side script; legacy maintenance only

2 **Web Forms**
.aspx event lifecycle; not part of ASP.NET Core

3 **MVC 5**
System.Web on .NET Framework

4 **ASP.NET Core MVC**
Microsoft.AspNetCore.* packages

K1 / A1

Full procedures: Learner Guide

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ASP.NET technology family boundary · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Classic ASP	Server-side script; legacy maintenance only	Where is this enforced?
Web Forms	.aspx event lifecycle; not part of ASP.NET Core	What is observable?
MVC 5	System.Web on .NET Framework	Who owns the value?
ASP.NET Core MVC	Microsoft.AspNetCore.* packages	What rejects invalid input?
Target	net10.0 for this course	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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ASP.NET technology family boundary · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Wrong tutorial	System.Web examples do not compile on Core
Wrong file	.aspx controls are not Razor
Wrong runtime	.NET Framework is Windows-only
Wrong host	IIS concepts are mistaken for the app pipeline

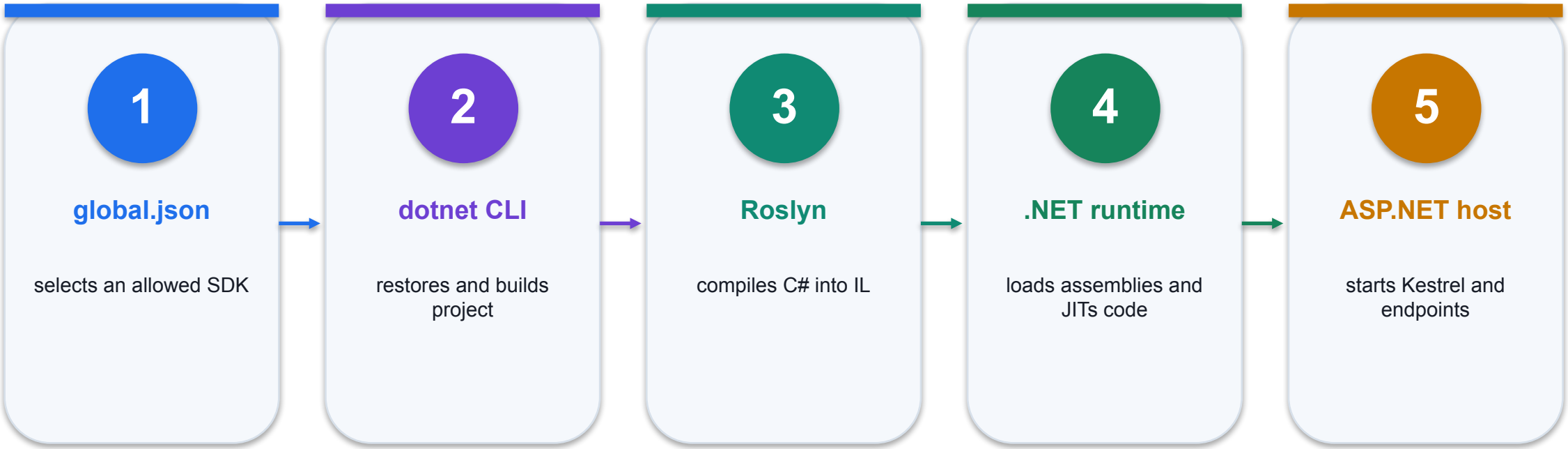
ACCEPTANCE EVIDENCE

- ✓ dotnet --list-sdks includes 10.0
- ✓ csproj targets net10.0
- ✓ No System.Web dependency exists

Record URL · command · status · build SHA

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SDK, runtime and target framework



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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SDK, runtime and target framework · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 {
2   "sdk": { "version": "10.0.100", "rollForward":
  "latestPatch" }
3 }
```

1 **SDK**
build tools plus templates

2 **Runtime**
executes a built application

3 **TFM**
net10.0 API compatibility contract

4 **RID**
optional platform-specific publish target

K1 / A1

Full procedures: Learner Guide

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TOPIC 1 · DESIGN DETAIL

SDK, runtime and target framework · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
SDK	build tools plus templates	Where is this enforced?
Runtime	executes a built application	What is observable?
TFM	net10.0 API compatibility contract	Who owns the value?
RID	optional platform-specific publish target	What rejects invalid input?
Lock	global.json prevents silent SDK drift	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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SDK, runtime and target framework · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

NETSDK1045	installed SDK cannot target net10.0
Restore error	package source or version is unavailable
Works on mine	SDK version was not pinned
Runtime missing	framework-dependent publish reached an empty host

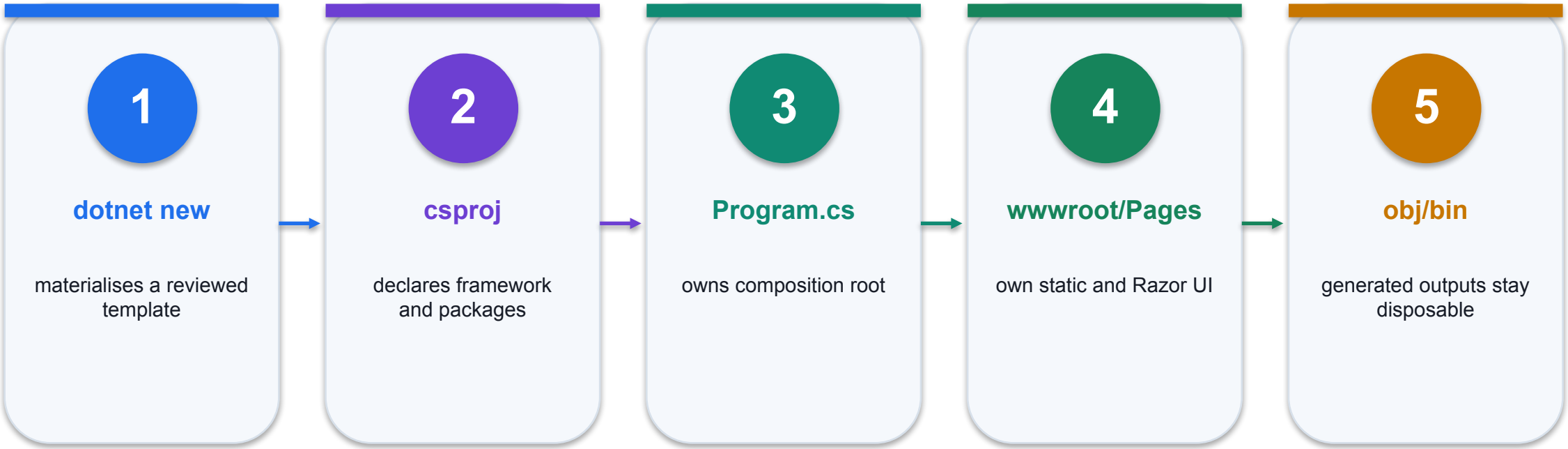
ACCEPTANCE EVIDENCE

- ✓ dotnet --info records SDK and runtime
- ✓ dotnet build ends with 0 errors
- ✓ global.json is committed with the design

Record URL · command · status · build SHA

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Project template and file ownership



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Project template and file ownership · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 dotnet new webapp -n LocalCRM --framework net10.0
2 dotnet new list web
```

1 **webapp**
Razor Pages project template

2 **webapi**
HTTP API template

3 **--no-https**
only for constrained local tests

4 **--use-controllers**
selects controller API style

K1 / A1

Full procedures: Learner Guide

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TOPIC 1 · DESIGN DETAIL

Project template and file ownership · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
webapp	Razor Pages project template	Where is this enforced?
webapi	HTTP API template	What is observable?
--no-https	only for constrained local tests	Who owns the value?
--use-controllers	selects controller API style	What rejects invalid input?
--dry-run	previews files before creation	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Project template and file ownership · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Template absent	required SDK/workload is missing
Generated files edited	obj/bin were mistaken for source
Secret committed	local settings were treated as portable
Name collision	namespace and folder identity diverged

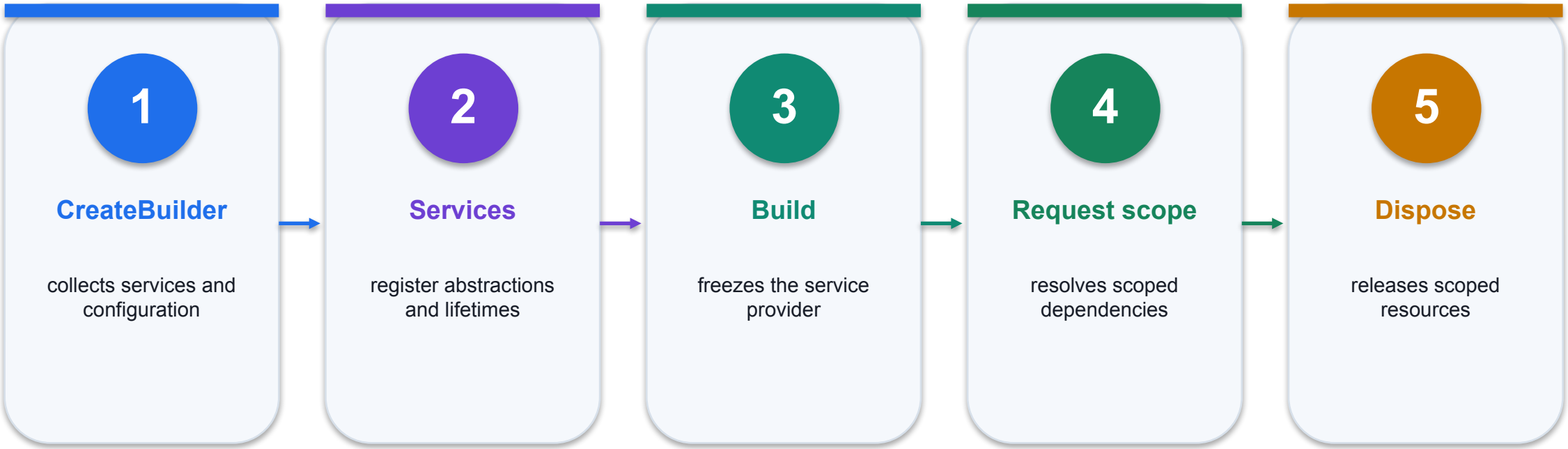
ACCEPTANCE EVIDENCE

- ✓ dotnet new --dry-run shows the intended template
- ✓ git status excludes bin/ and obj/
- ✓ dotnet run exposes the generated endpoint

Record URL · command · status · build SHA

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Composition root and dependency injection



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Composition root and dependency injection · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  builder.Services.AddScoped<IContactStore,
   SqliteContactStore>();
2  var app = builder.Build();
```

1. **Singleton**
   one instance for process lifetime
2. **Scoped**
   one instance per HTTP request
3. **Transient**
   new instance per resolution
4. **Constructor**
   declares required dependency

K1 / A1

Full procedures: Learner Guide

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Composition root and dependency injection · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Singleton	one instance for process lifetime	Where is this enforced?
Scoped	one instance per HTTP request	What is observable?
Transient	new instance per resolution	Who owns the value?
Constructor	declares required dependency	What rejects invalid input?
Interface	separates consumer from implementation	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Composition root and dependency injection · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Cannot resolve	service was never registered
Captive dependency	singleton holds a scoped service
Service locator	runtime GetService hides the contract
Duplicate registration	last registration silently wins

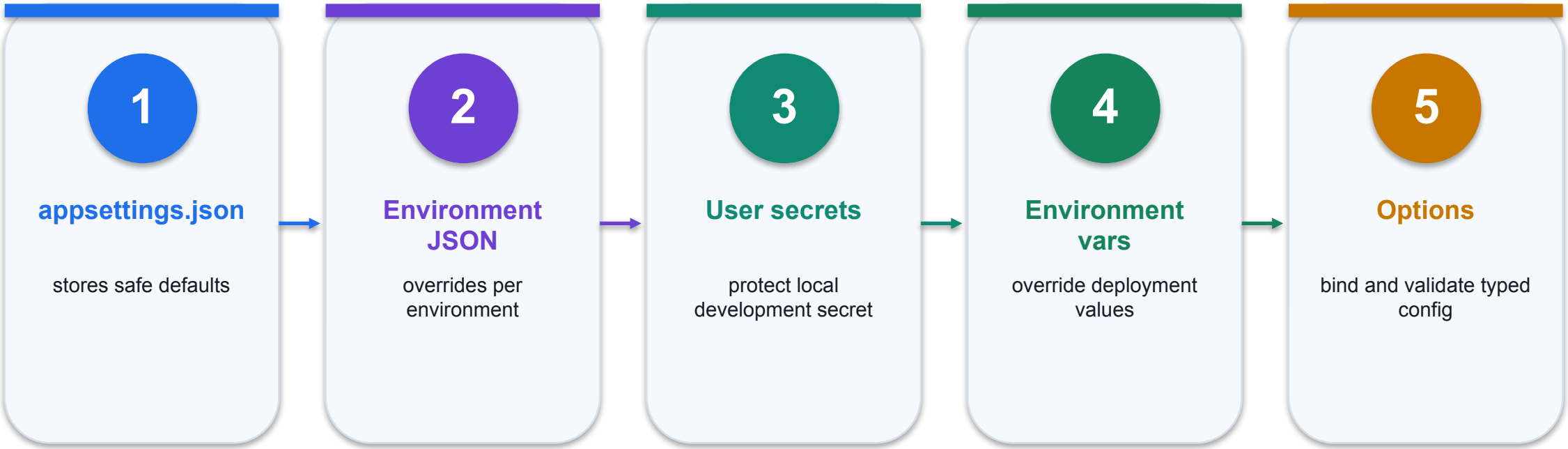
ACCEPTANCE EVIDENCE

- ✓ Startup completes without DI exception
- ✓ One scoped instance is reused within a request
- ✓ Unit test can replace IContactStore

Record URL · command · status · build SHA

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Configuration provider precedence



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Configuration provider precedence · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 builder.Services.AddOptions<AppOptions>()
2   .BindConfiguration("App")
3   .ValidateDataAnnotations().ValidateOnStart();
```

1 **Key path**

ConnectionStrings:LocalCRM

2 **Env encoding**

double underscore represents colon

3 **Secret**

never committed to appsettings.json

4 **Options**

ValidateOnStart catches invalid config

K1 / A1

Full procedures: Learner Guide

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Configuration provider precedence · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Key path	ConnectionStrings:LocalCRM	Where is this enforced?
Env encoding	double underscore represents colon	What is observable?
Secret	never committed to appsettings.json	Who owns the value?
Options	ValidateOnStart catches invalid config	What rejects invalid input?
Precedence	later provider overrides earlier	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Configuration provider precedence · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Null option	key path does not match property
Dev leak	secret landed in source control
Wrong environment	ASPNETCORE_ENVIRONMENT is incorrect
Late failure	options were not validated at startup

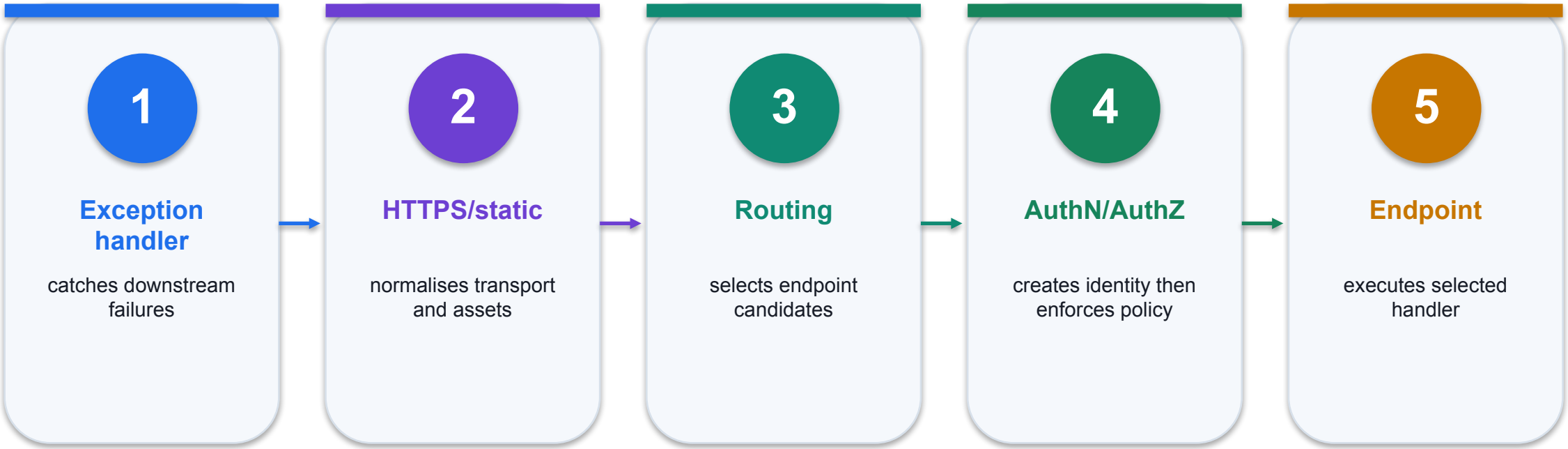
ACCEPTANCE EVIDENCE

- ✓ Startup rejects an invalid required setting
- ✓ env var changes the resolved value
- ✓ secret scan finds no credential

Record URL · command · status · build SHA

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Middleware pipeline ordering



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Middleware pipeline ordering · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 app.UseExceptionHandler("/Error");
2 app.UseHttpsRedirection();
3 app.UseAuthorization();
4 app.MapRazorPages();
```

1. **Use**
   can run before and after next
2. **Run**
   terminal middleware
3. **Map**
   branches by path
4. **Order**
   registration order is execution order

K1 / A1

Full procedures: Learner Guide

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TOPIC 1 · DESIGN DETAIL

Middleware pipeline ordering · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Use	can run before and after next	Where is this enforced?
Run	terminal middleware	What is observable?
Map	branches by path	Who owns the value?
Order	registration order is execution order	What rejects invalid input?
Short-circuit	response ends without downstream call	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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TOPIC 1 · DIAGNOSE + VERIFY

Middleware pipeline ordering · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

401/403 surprise	authorization ran before authentication
Headers absent	middleware ran after endpoint
Static 404	static files were not mapped
Details leak	developer exception page reached production

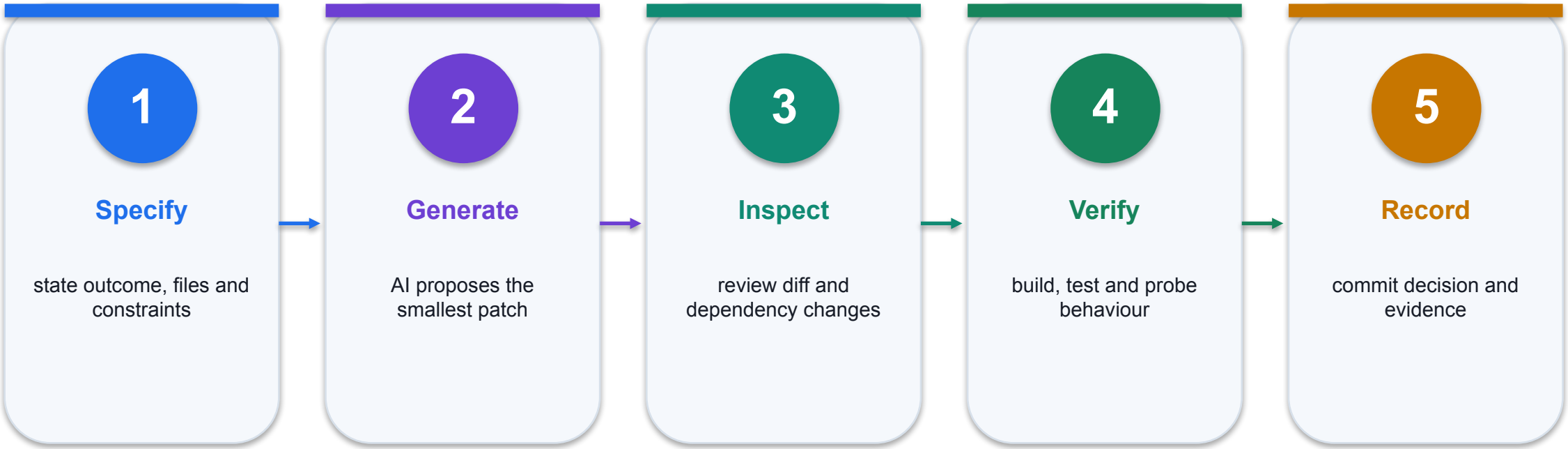
ACCEPTANCE EVIDENCE

- ✓ Trace log shows expected middleware order
- ✓ Production errors use generic ProblemDetails
- ✓ Protected endpoint enforces its policy

Record URL · command · status · build SHA

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Vibe coding control loop



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Vibe coding control loop · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 Prompt: Add GET /health to Program.cs only.
2 Constraints: net10.0; no new packages.
3 Verify: dotnet test && curl -i /health
```

1 **Context**
target files and accepted architecture

2 **Constraints**
net10.0, no secrets, accessible UI

3 **Acceptance**
commands and exact observable result

4 **Diff budget**
reject unrelated changes

K1 / A1

Full procedures: Learner Guide

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Vibe coding control loop · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Context	target files and accepted architecture	Where is this enforced?
Constraints	net10.0, no secrets, accessible UI	What is observable?
Acceptance	commands and exact observable result	Who owns the value?
Diff budget	reject unrelated changes	What rejects invalid input?
Rollback	checkpoint before generation	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Vibe coding control loop · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Prompt drift	scope did not name files
Hallucinated API	package/version was not checked
Silent regression	only the happy path was probed
Mega-diff	one prompt mixed unrelated changes

ACCEPTANCE EVIDENCE

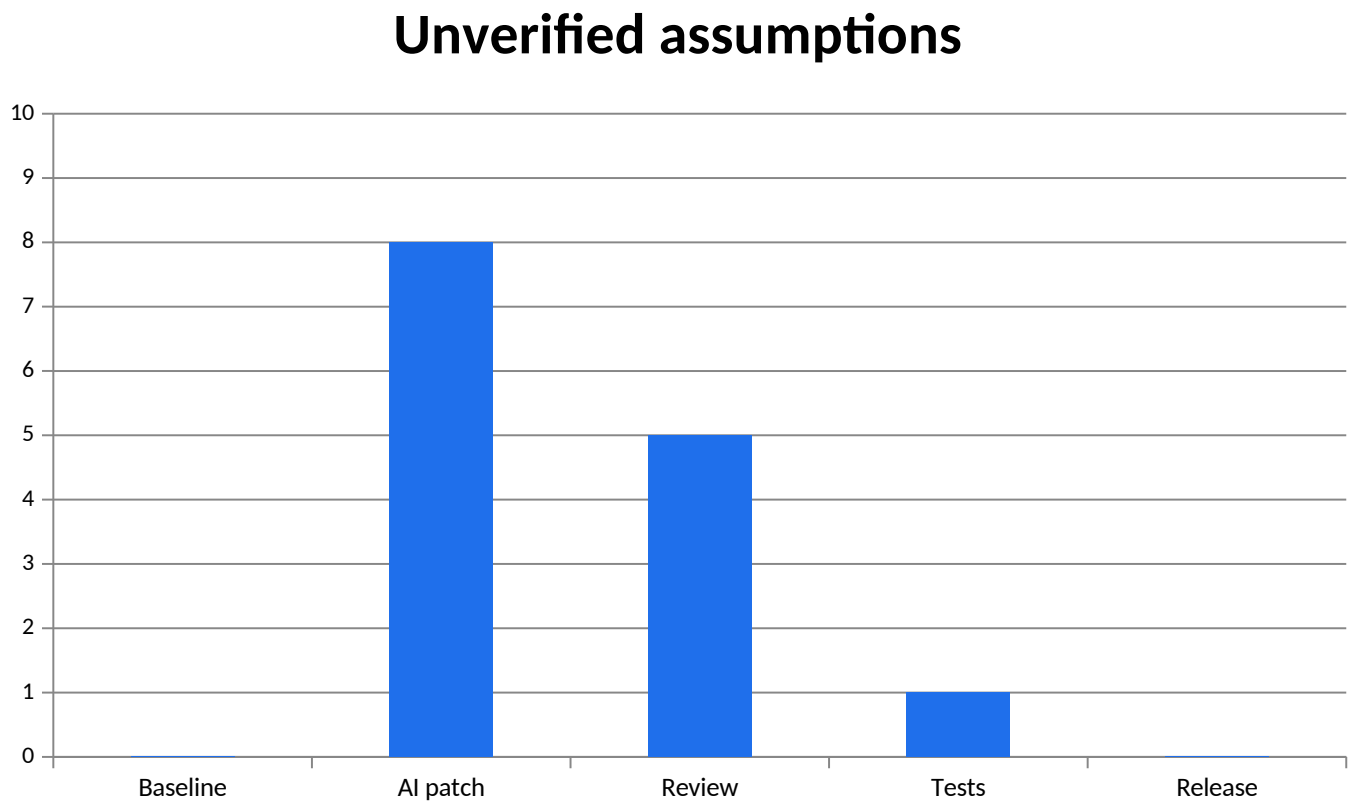
- ✓ git diff contains only declared files
- ✓ dotnet test is green
- ✓ manual probe produces named evidence

Record URL · command · status · build SHA

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TOPIC 1 · EDITABLE NATIVE CHART

LO1 evidence profile



WHAT THE DATA SHOWS

The review loop drives assumptions to zero before release.

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

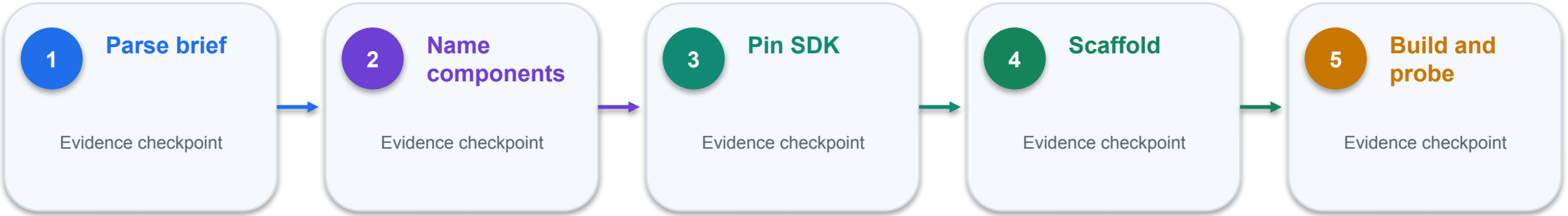
EDITABLE CHART

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55 MIN

Activity 1: Translate Requirements into a Verified ASP.NET Core Scaffold

Turn a LocalCRM customer-contact brief into component boundaries and a reproducible .NET 10 Razor Pages project.



YOU'LL PRODUCE **requirements.md, component-map.md, pinned SDK file and a clean net10.0 Razor Pages scaffold**

Local snapshot: activities/activity-01-requirements-and-scaffold/src/LocalCRM · full commands in the Learner Guide

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Activity 1 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Act as a software-design assistant. Convert the supplied LocalCRM brief into UI, application-service, data and hosting components. Preserve requirement IDs. Do not write code yet. Return assumptions separately.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 1 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: requirements.md, component-map.md, pinned SDK file and a clean net10.0 Razor Pages scaffold.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

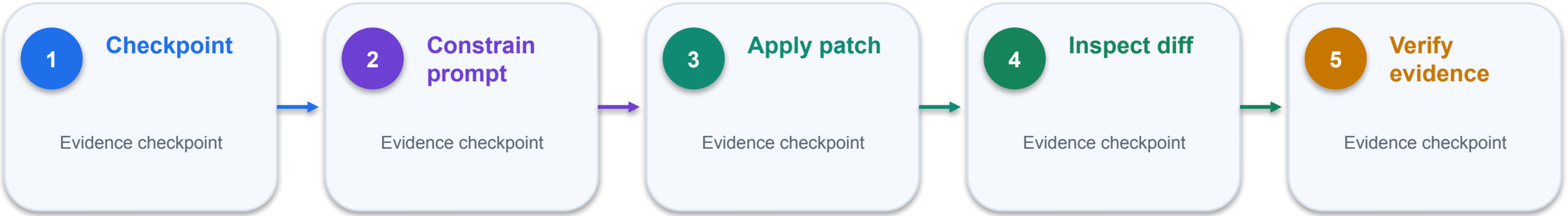
K1/A1 EVIDENCE

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45 MIN

Activity 2: Run a Controlled Vibe-Coding Change

Add a health endpoint through a specify–generate–inspect–verify loop without accepting unrelated AI edits.



YOU'LL PRODUCE **a reviewed /health endpoint, build evidence and a decision log**

Local snapshot: activities/activity-02-controlled-vibe-change/src/LocalCRM · full commands in the Learner Guide

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Activity 2 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Modify LocalCRM/Program.cs only. Add GET /health returning HTTP 200 JSON { status: 'ok', service: 'LocalCRM' }. Do not add packages or reorder existing middleware. Explain the exact diff and provide curl acceptance evidence.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 2 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: a reviewed /health endpoint, build evidence and a decision log.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

K1/A1 EVIDENCE

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LO1 · K1 / A1

Topic 1 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.

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TOPIC 2 · LO2 · K2 / A2

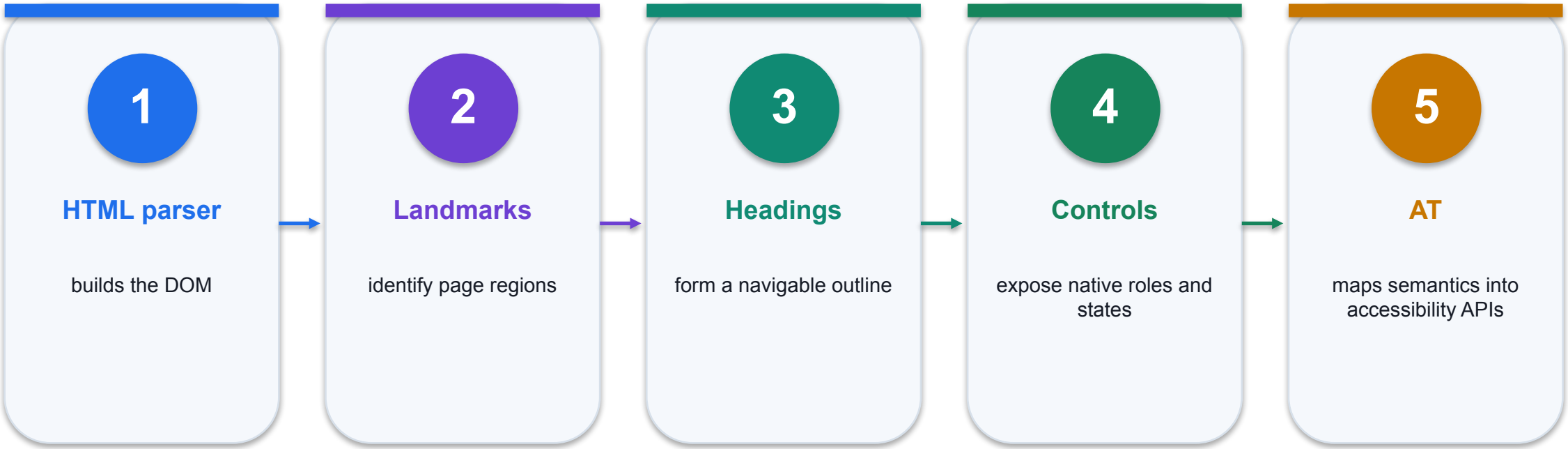
02

Accessible UI and Organisational Vibe-Coding Practice

semantic structure → accessible interaction → reviewable AI-assisted change

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Semantic document structure



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Semantic document structure · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <main>
2   <h1>CRM contacts</h1>
3   <button type="submit">Add contact</button>
4 </main>
```

1 **main**
one primary content region

2 **nav**
named navigation collection

3 **h1–h6**
do not skip levels for styling

4 **button**
action with keyboard behaviour

K2 / A2

Full procedures: Learner Guide

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TOPIC 2 · DESIGN DETAIL

Semantic document structure · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
main	one primary content region	Where is this enforced?
nav	named navigation collection	What is observable?
h1–h6	do not skip levels for styling	Who owns the value?
button	action with keyboard behaviour	What rejects invalid input?
a[href]	navigation to a real target	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Semantic document structure · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Div soup	visual boxes have no programmatic role
Heading jump	outline no longer communicates hierarchy
Clickable span	keyboard activation is missing
Duplicate main	landmark navigation becomes ambiguous

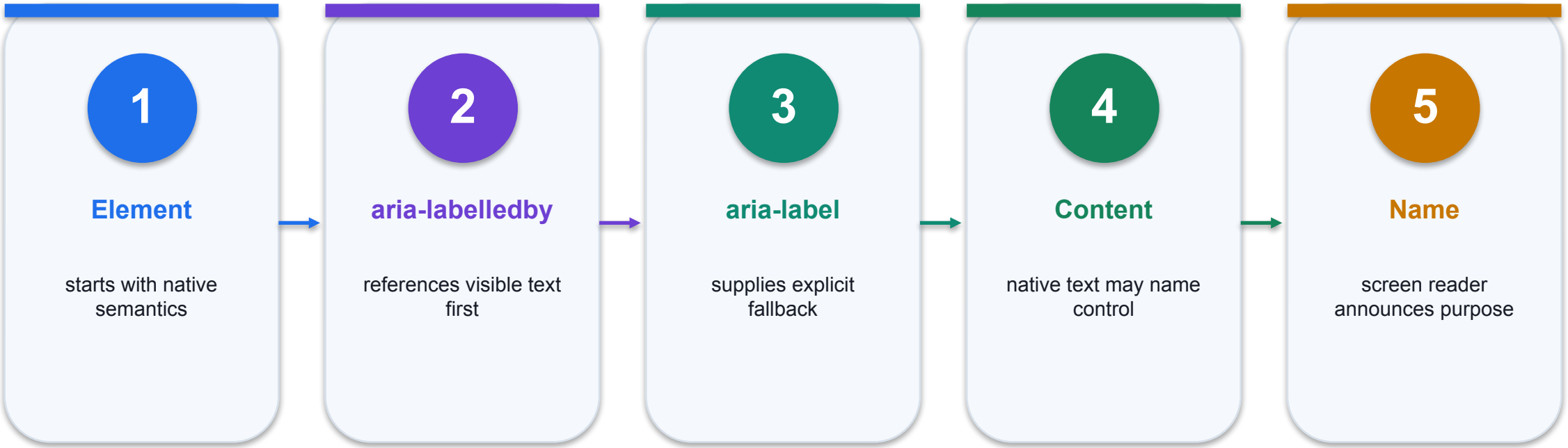
ACCEPTANCE EVIDENCE

- ✓ Accessibility tree exposes main and button
- ✓ Heading outline has one h1
- ✓ Tab and Enter activate the control

Record URL · command · status · build SHA

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Accessible name computation



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Accessible name computation · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <label for="email">Summary</label>
2 <input id="email" name="email" type="email" aria-
  describedby="email-help">
```

1 **label for**
associates text with an input id

2 **alt**
names an informative image

3 **empty alt**
marks decorative image

4 **aria-describedby**
adds help, not the name

K2 / A2

Full procedures: Learner Guide

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TOPIC 2 · DESIGN DETAIL

Accessible name computation · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
label for	associates text with an input id	Where is this enforced?
alt	names an informative image	What is observable?
empty alt	marks decorative image	Who owns the value?
aria-describedby	adds help, not the name	What rejects invalid input?
Unique id	prevents broken reference	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Accessible name computation · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Placeholder only	name disappears after typing
Filename announced	image has missing or poor alt
Duplicate id	label binds to the wrong control
ARIA override	visible and spoken names disagree

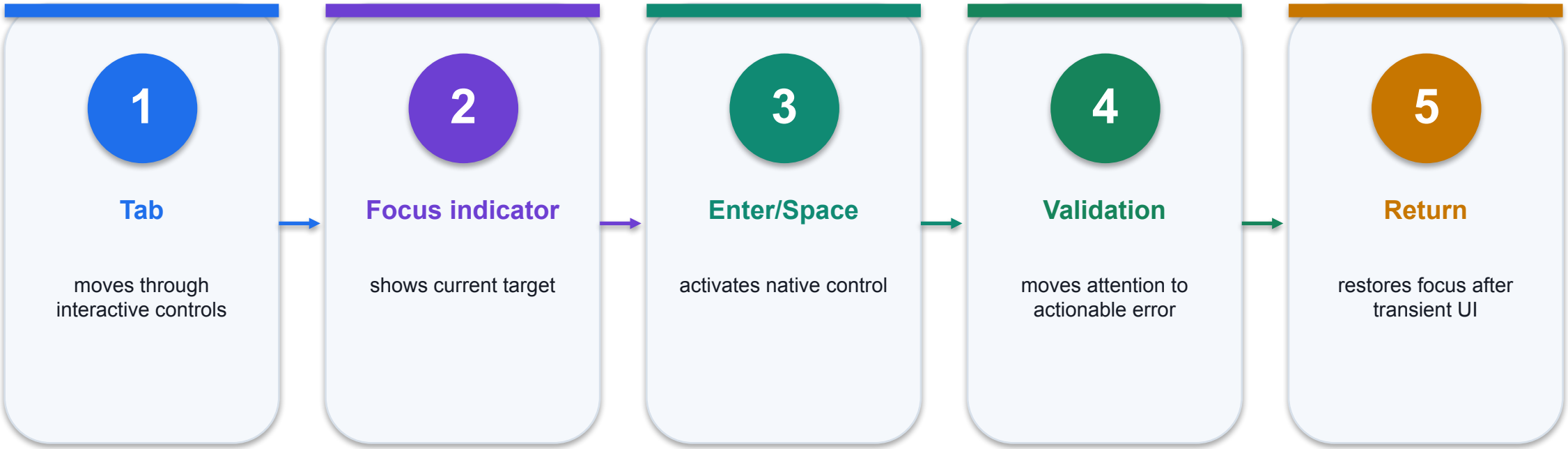
ACCEPTANCE EVIDENCE

- ✓ Browser Accessibility pane shows the intended name
- ✓ Screen reader announces label and help
- ✓ axe reports no form-label violation

Record URL · command · status · build SHA

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Keyboard and focus contract



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Keyboard and focus contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  button:focus-visible { outline: 3px solid #1f6feb; outline-
   offset: 3px; }
```

1 **Order**
DOM order should match visual order

2 **tabindex=0**
adds custom widget to natural order

3 **tabindex=-1**
allows scripted focus only

4 **:focus-visible**
keeps visible keyboard cue

K2 / A2

Full procedures: Learner Guide

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Keyboard and focus contract · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Order	DOM order should match visual order	Where is this enforced?
tabindex=0	adds custom widget to natural order	What is observable?
tabindex=-1	allows scripted focus only	Who owns the value?
:focus-visible	keeps visible keyboard cue	What rejects invalid input?
Escape	closes modal/menu and restores focus	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Keyboard and focus contract · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Focus trap	user cannot leave a component
Focus loss	removed node held focus
Positive tabindex	manual ordering becomes brittle
Mouse-only handler	keyboard event path is absent

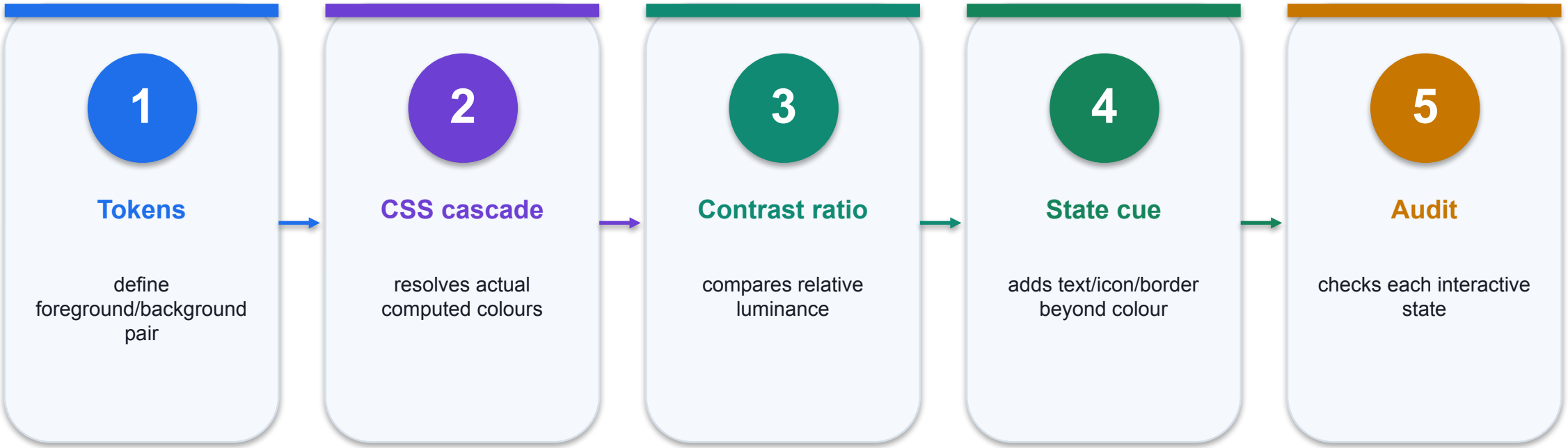
ACCEPTANCE EVIDENCE

- ✓ Entire workflow completes without mouse
- ✓ Focus indicator remains visible
- ✓ Modal Escape returns focus to opener

Record URL · command · status · build SHA

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Contrast and non-colour cues



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Contrast and non-colour cues · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  .error { color:#b42318; border-left:4px solid #b42318; }
2  .error::before { content:"Error: "; font-weight:700; }
```

1 **Normal text**
target 4.5:1

2 **Large text**
target 3:1

3 **UI component**
target 3:1 against adjacent colour

4 **Link**
not identified by colour alone

K2 / A2

Full procedures: Learner Guide

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Contrast and non-colour cues · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Normal text	target 4.5:1	Where is this enforced?
Large text	target 3:1	What is observable?
UI component	target 3:1 against adjacent colour	Who owns the value?
Link	not identified by colour alone	What rejects invalid input?
State	hover, focus, disabled and error all checked	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Contrast and non-colour cues · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Token passes only once	hover state drops contrast
Red/green only	colour vision users lose status
Text on image	background changes under glyphs
Disabled too faint	essential text becomes unreadable

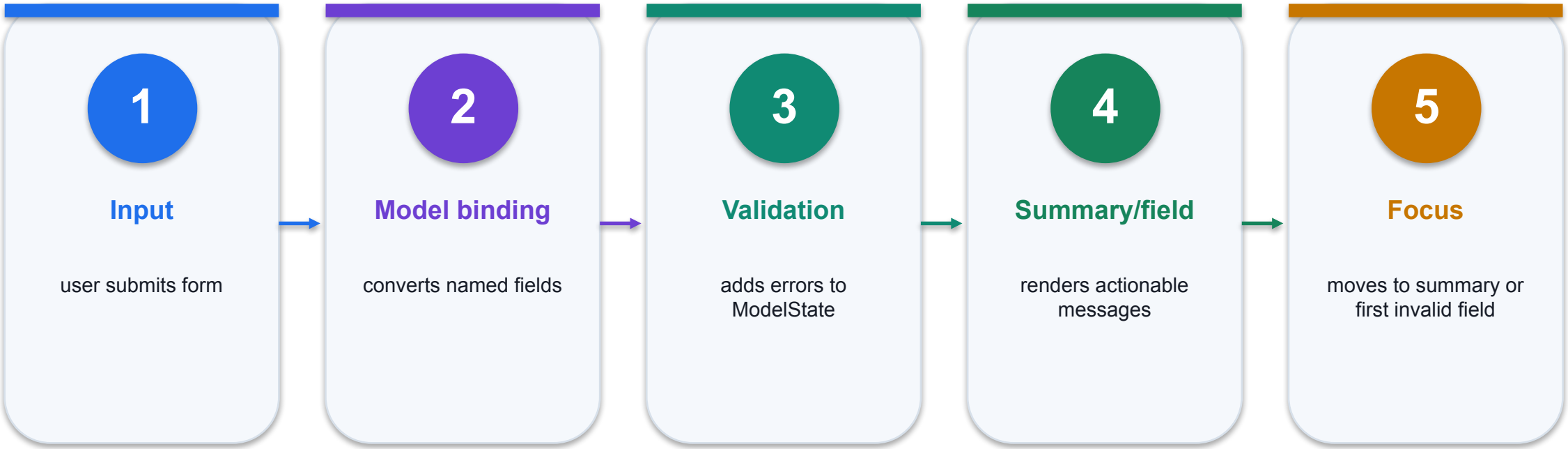
ACCEPTANCE EVIDENCE

- ✓ Contrast checker records each token pair
- ✓ Status remains understandable in grayscale
- ✓ Lighthouse/axe result is attached

Record URL · command · status · build SHA

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Validation error experience



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Validation error experience · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <span asp-validation-for="Input.Email"></span>
2 <div asp-validation-summary="ModelOnly" tabindex="-1"></div>
```

1. **asp-validation-for**
   field-specific message
2. **asp-validation-summary**
   page-level error collection
3. **aria-invalid**
   marks failing input
4. **describedby**
   associates message

K2 / A2

Full procedures: Learner Guide

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Validation error experience · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
asp-validation-for	field-specific message	Where is this enforced?
asp-validation-summary	page-level error collection	What is observable?
aria-invalid	marks failing input	Who owns the value?
describedby	associates message	What rejects invalid input?
Server check	remains authoritative without JavaScript	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Validation error experience · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Client-only	bypassed request accepts invalid data
Generic error	user cannot identify repair
Focus unchanged	screen-reader user misses result
Values lost	failed post rebuilds an empty form

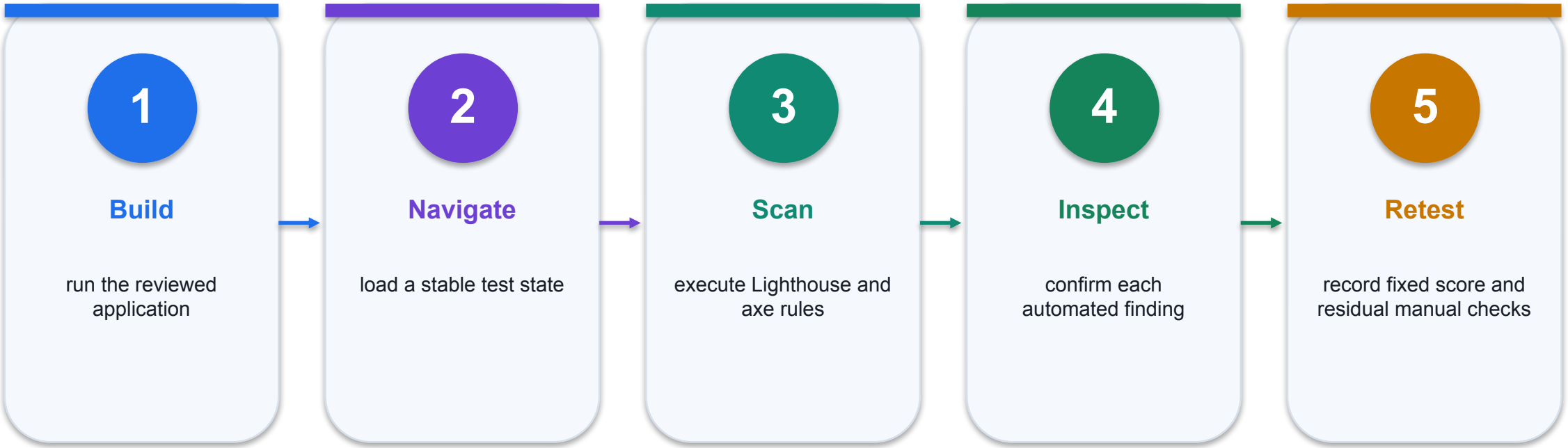
ACCEPTANCE EVIDENCE

- ✓ Empty submit shows specific message
- ✓ Error is associated with the field
- ✓ Server rejects the same invalid payload

Record URL · command · status · build SHA

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Lighthouse and axe evidence workflow



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Lighthouse and axe evidence workflow · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 npx @axe-core/cli http://localhost:5000 --exit
2 # Expected: 0 violations (manual checks still required)
```

1 **Automated**
finds detectable rule violations

2 **Manual**
keyboard, reading order and meaning

3 **Baseline**
store score and violation IDs

4 **Regression**
fail build on new serious violations

K2 / A2

Full procedures: Learner Guide

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Lighthouse and axe evidence workflow · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Automated	finds detectable rule violations	Where is this enforced?
Manual	keyboard, reading order and meaning	What is observable?
Baseline	store score and violation IDs	Who owns the value?
Regression	fail build on new serious violations	What rejects invalid input?
Evidence	HTML/JSON report plus screenshot	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Lighthouse and axe evidence workflow · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

False confidence	score was treated as full conformance
Flaky state	scan ran before page settled
Third-party noise	scope was not documented
No baseline	regression cannot be distinguished

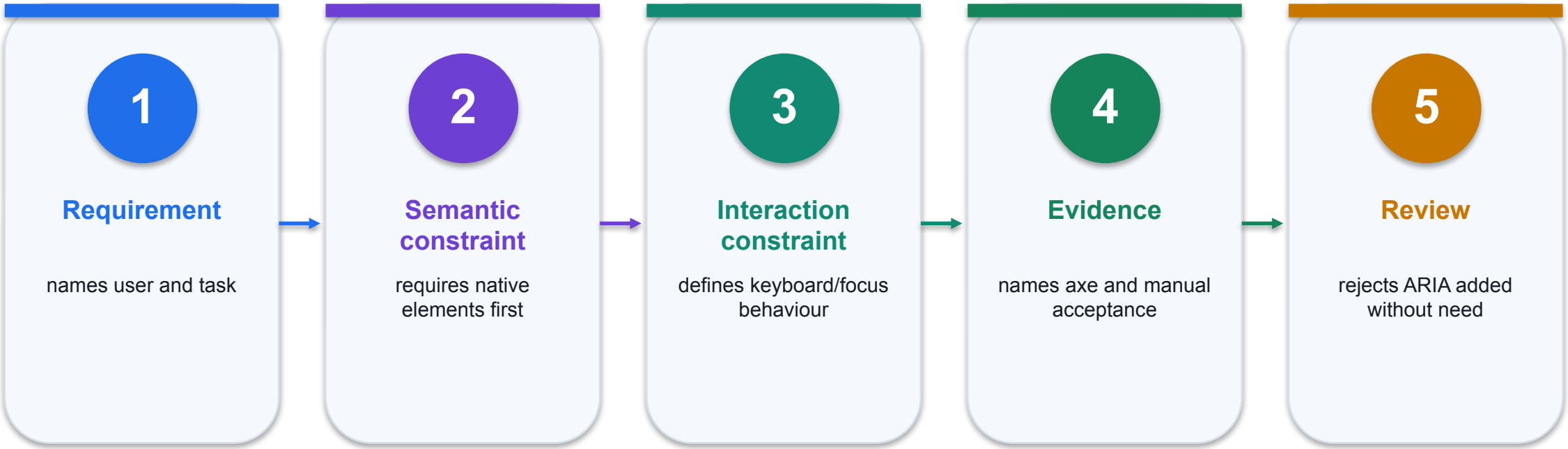
ACCEPTANCE EVIDENCE

- ✓ axe exits 0 for the target state
- ✓ Manual keyboard checklist is signed
- ✓ Report names URL, date and build SHA

Record URL · command · status · build SHA

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Prompt constraints for accessible output



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Prompt constraints for accessible output · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 Prompt: Make Pages/Contacts/Create.cshtml keyboard complete.
2 Use native controls; preserve label text.
3 Acceptance: axe=0 and Tab→Enter submits.
```

1 **Files**
limit AI to the intended Razor/CSS files

2 **Standard**
reference WCAG outcome, not vague accessibility

3 **Prohibition**
no clickable divs or positive tabindex

4 **Acceptance**
keyboard path plus automated scan

K2 / A2

Full procedures: Learner Guide

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Prompt constraints for accessible output · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Files	limit AI to the intended Razor/CSS files	Where is this enforced?
Standard	reference WCAG outcome, not vague accessibility	What is observable?
Prohibition	no clickable divs or positive tabindex	Who owns the value?
Acceptance	keyboard path plus automated scan	What rejects invalid input?
Output	request patch and rationale, not prose only	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Prompt constraints for accessible output · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Vague prompt	model optimises appearance only
ARIA cargo cult	redundant roles conflict with native semantics
Hidden focus	CSS reset removes outline
Untested claim	AI says accessible without evidence

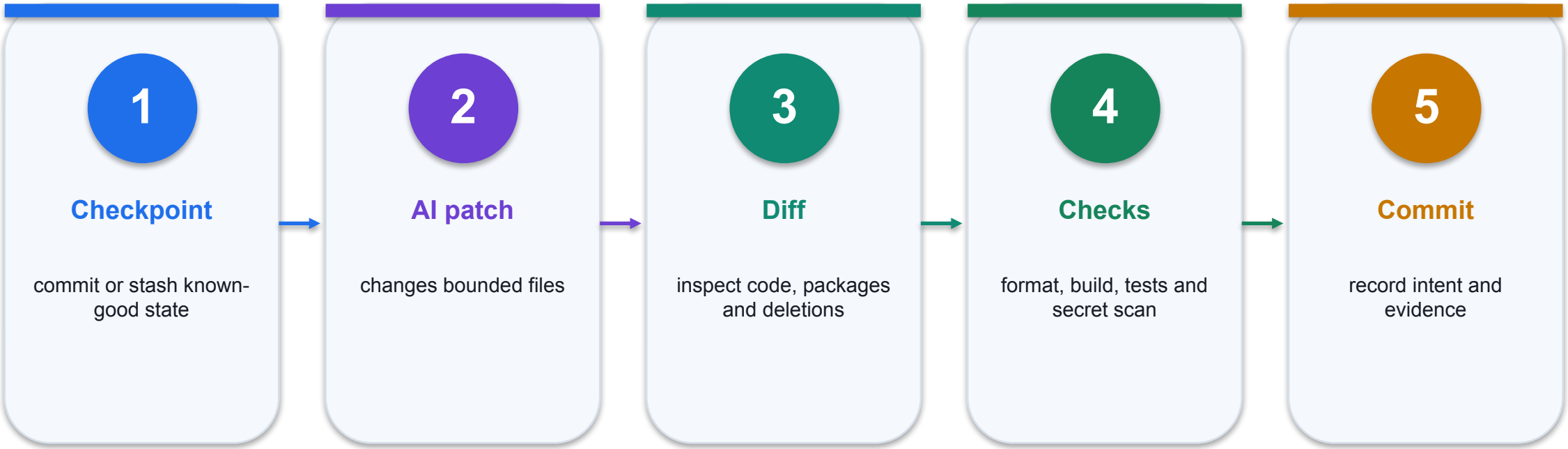
ACCEPTANCE EVIDENCE

- ✓ Diff uses native elements
- ✓ Tab order matches DOM order
- ✓ Acceptance evidence is reproducible

Record URL · command · status · build SHA

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Git diff and review guardrails



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Git diff and review guardrails · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 git diff --check
2 dotnet format --verify-no-changes
3 dotnet test
4 git diff --name-only
```

1 **git diff --stat**
detects unexpected breadth

2 **git diff --check**
finds whitespace defects

3 **dotnet format --verify-no-changes**
enforces style

4 **dotnet test**
proves behavioural contract

K2 / A2

Full procedures: Learner Guide

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Git diff and review guardrails · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
git diff --stat	detects unexpected breadth	Where is this enforced?
git diff --check	finds whitespace defects	What is observable?
dotnet format --verify-no-changes	enforces style	Who owns the value?
dotnet test	proves behavioural contract	What rejects invalid input?
Secret scan	blocks credentials before history	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Git diff and review guardrails · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Generated noise	bin/obj entered the diff
Package surprise	AI added dependency without review
Secret leak	connection string was committed
Mixed commit	unrelated changes obscure review

ACCEPTANCE EVIDENCE

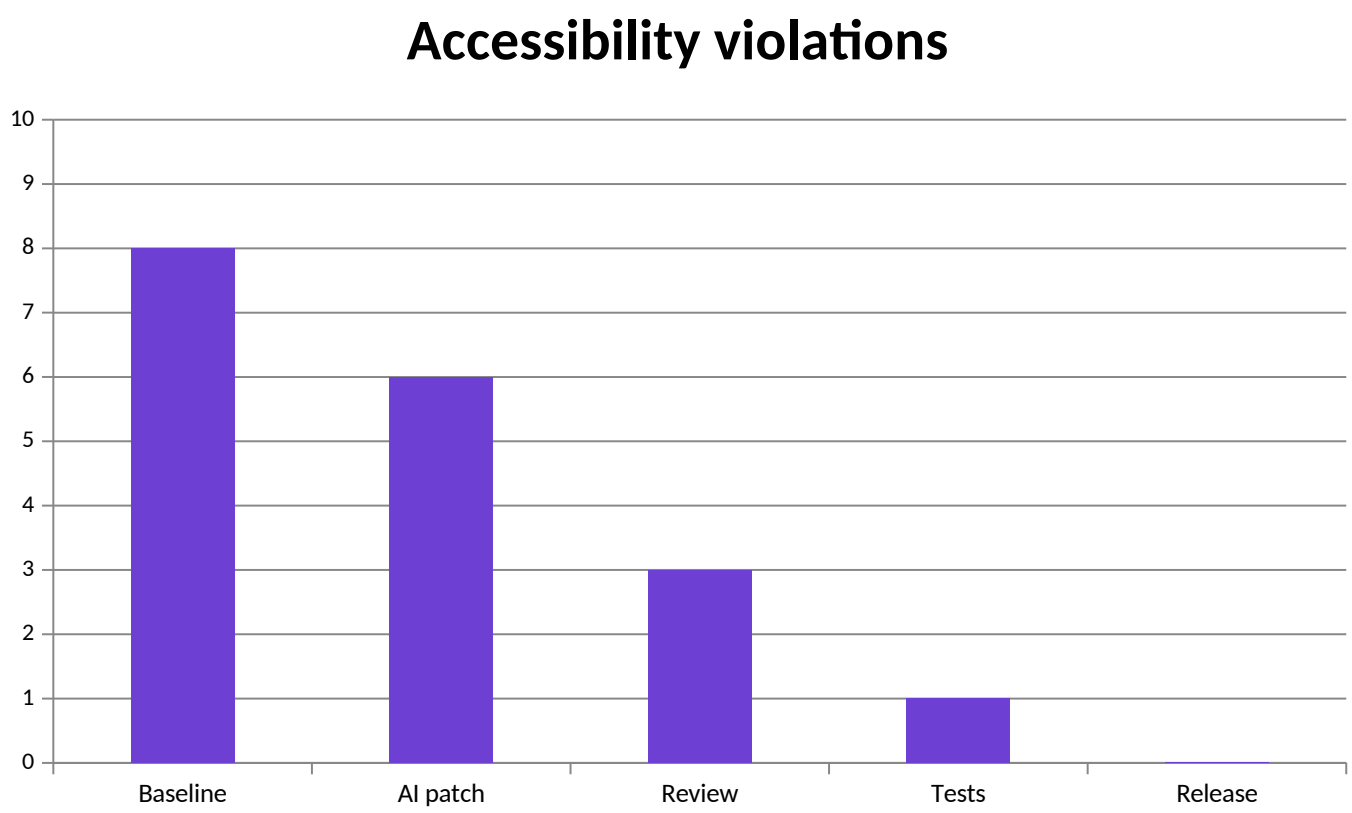
- ✓ Only scoped source files changed
- ✓ All local gates exit 0
- ✓ Commit message links requirement and evidence

Record URL · command · status · build SHA

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TOPIC 2 · EDITABLE NATIVE CHART

LO2 evidence profile



WHAT THE DATA SHOWS

Automated evidence improves; manual keyboard checks still remain mandatory.

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

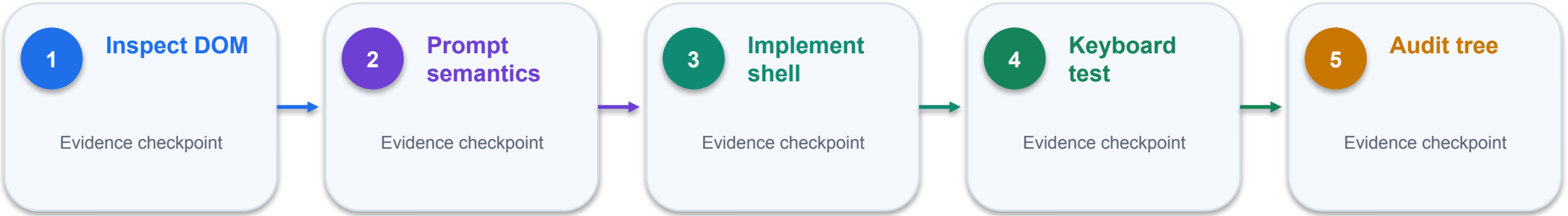
EDITABLE CHART

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55 MIN

Activity 3: Build an Accessible Razor Pages Shell

Create semantic navigation, visible keyboard focus and a labelled customer-contact entry point.



YOU'LL PRODUCE **an accessible layout and Contacts index page with a reproducible keyboard checklist**

Local snapshot: activities/activity-03-accessible-razor-shell/src/LocalCRM · full commands in the Learner Guide

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Activity 3 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Update the Razor layout and add Pages/Contacts/Index.cshtml. Use native header/nav/main/footer landmarks, one h1, descriptive link text and visible :focus-visible styles. No clickable divs, positive tabindex or ARIA when native HTML suffices.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 3 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: an accessible layout and Contacts index page with a reproducible keyboard checklist.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

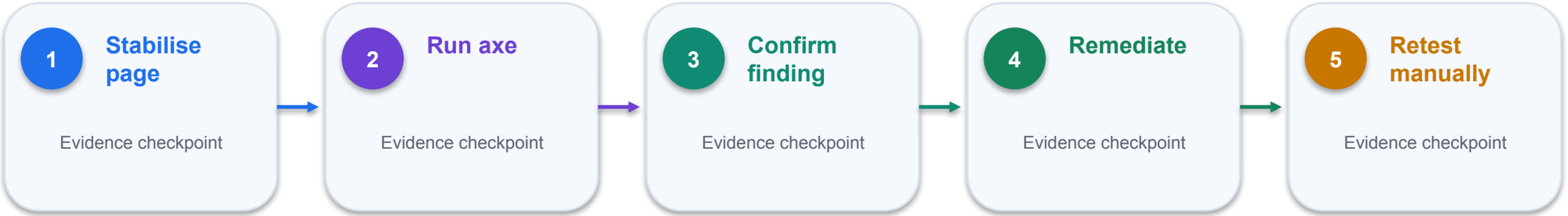
K2/A2 EVIDENCE

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45 MIN

Activity 4: Audit Accessibility with axe and Manual Evidence

Create a repeatable automated-plus-manual accessibility gate for the running Razor Pages UI.



YOU'LL PRODUCE **axe JSON evidence, a manual keyboard checklist and a reviewed remediation diff**

Local snapshot: activities/activity-04-accessibility-audit/src/LocalCRM · full commands in the Learner Guide

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Activity 4 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Review the supplied axe findings and the Razor markup. Propose the smallest semantic fix for each confirmed violation. Do not suppress rules. State which outcomes still require manual keyboard or screen-reader checks.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 4 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: axe JSON evidence, a manual keyboard checklist and a reviewed remediation diff.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

K2/A2 EVIDENCE

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LO2 · K2 / A2

Topic 2 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.

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TOPIC 3 · LO3 · K3 / A3

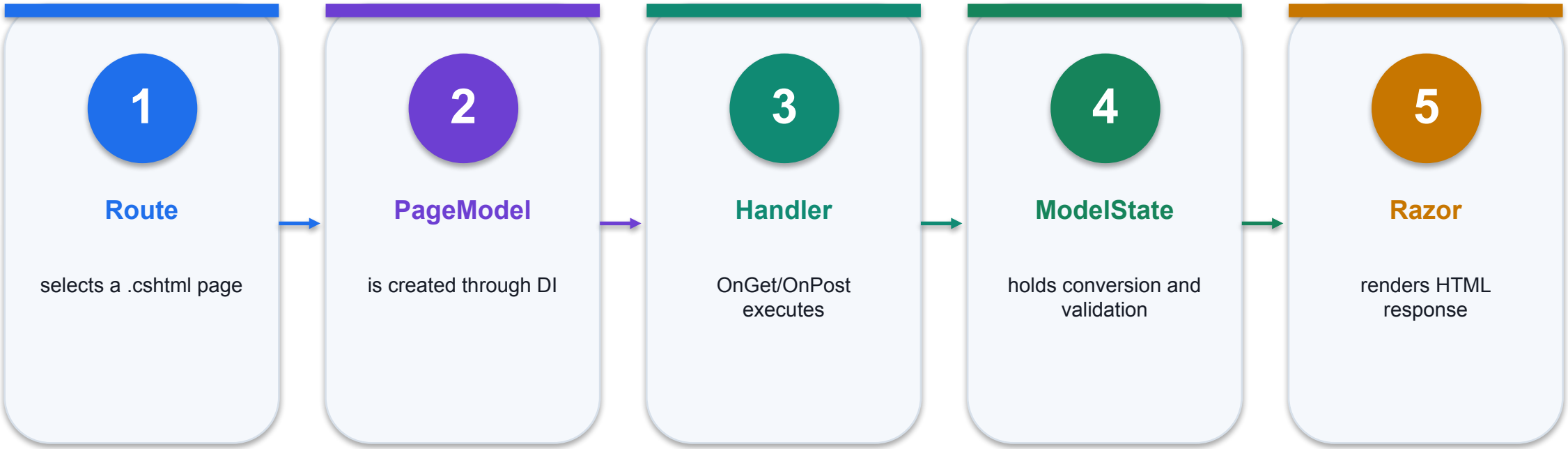
Razor Pages Controls, Binding and Persistence

03

page-focused UI → binding and validation → antiforgery → durable state

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Razor Page execution model



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Razor Page execution model · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 public class CreateModel(IContactStore store) : PageModel
2 { public void OnGet() { }
3   public async Task<IActionResult> OnPostAsync() {
  /*...*/ } }
```

1 **.cshtml**
markup plus directives and Tag Helpers

2 **.cshtml.cs**
page handlers and input model

3 **OnGet**
safe read path

4 **OnPost**
form mutation path

K3 / A3

Full procedures: Learner Guide

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Razor Page execution model · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
.cshtml	markup plus directives and Tag Helpers	Where is this enforced?
.cshtml.cs	page handlers and input model	What is observable?
OnGet	safe read path	Who owns the value?
OnPost	form mutation path	What rejects invalid input?
IActionResult	redirect, page or status result	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Razor Page execution model · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Handler not called	method/name convention mismatched
Null service	DI registration missing
Double submit	post returned Page instead of PRG redirect
Logic in view	separation of concerns collapsed

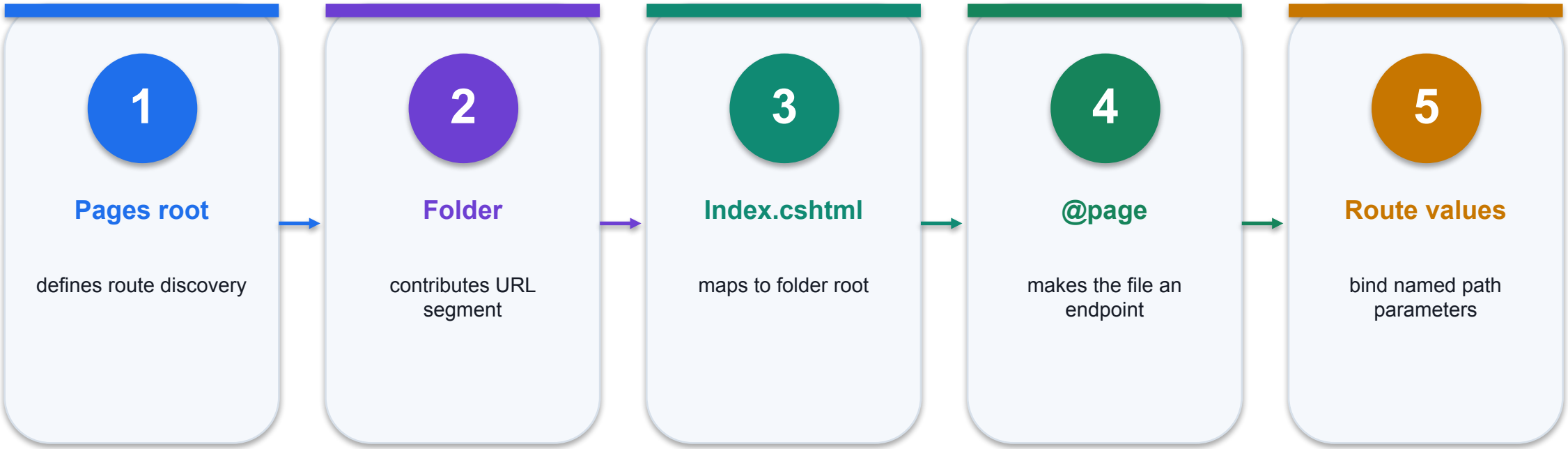
ACCEPTANCE EVIDENCE

- ✓ GET selects OnGet
- ✓ POST selects OnPostAsync
- ✓ Successful POST redirects to index

Record URL · command · status · build SHA

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Razor Pages routing conventions



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Razor Pages routing conventions · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 @page "{id:int}"
2 <a asp-page="./Details" asp-route-id="@item.Id">View</a>
```

1 **/Pages/Contacts/Index.cshtml**
route /Contacts

2 **@page "{id:int}"**
typed route constraint

3 **asp-page**
generates page URL

4 **asp-route-id**
adds route value

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Razor Pages routing conventions · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
/Pages/Contacts/ Index.cshtml	route /Contacts	Where is this enforced?
@page "{id:int}"	typed route constraint	What is observable?
asp-page	generates page URL	Who owns the value?
asp-route-id	adds route value	What rejects invalid input?
Areas	partition larger applications	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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TOPIC 3 · DIAGNOSE + VERIFY

Razor Pages routing conventions · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

404	@page is missing or path is wrong
Ambiguous link	relative page path resolves elsewhere
String id	route constraint was omitted
Hardcoded URL	renaming page breaks navigation

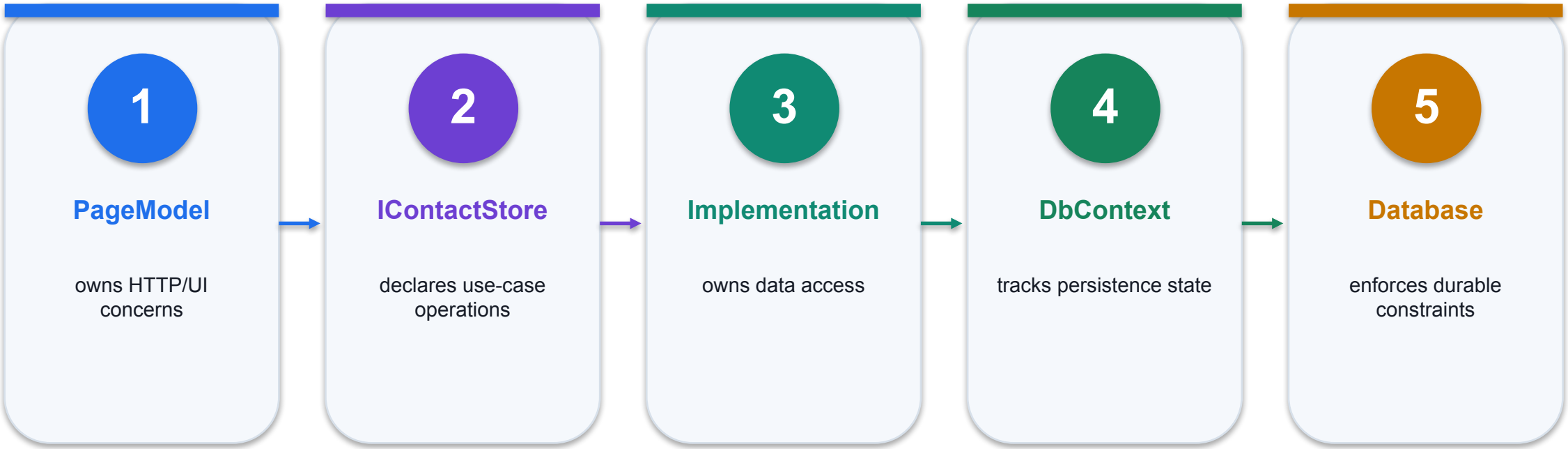
ACCEPTANCE EVIDENCE

- ✓ LinkGenerator produces /Contacts/42
- ✓ Non-integer id does not match
- ✓ Renaming folder requires no hardcoded href edit

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Service boundary behind the PageModel



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Service boundary behind the PageModel · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  public interface IContactStore
2  { Task<int> AddAsync(NewContact input, CancellationToken ct);
3    Task<IReadOnlyList<ContactRow>> ListAsync(CancellationToken
   ct); }
```

1 Interface

Task<List<CustomerContact>> ListAsync

2 Cancellation

accept RequestAborted token

3 DTO/ViewModel

limits exposed fields

4 Scoped lifetime

one DbContext per request

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Service boundary behind the PageModel · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Interface	Task<List<CustomerContact>> ListAsync	Where is this enforced?
Cancellation	accept RequestAborted token	What is observable?
DTO/ViewModel	limits exposed fields	Who owns the value?
Scoped lifetime	one DbContext per request	What rejects invalid input?
No static state	supports concurrent requests	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Service boundary behind the PageModel · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Fat PageModel	SQL and presentation logic are mixed
Static list	data races and restart loss
Sync-over-async	.Result can block request threads
Leaky entity	persistence shape becomes public contract

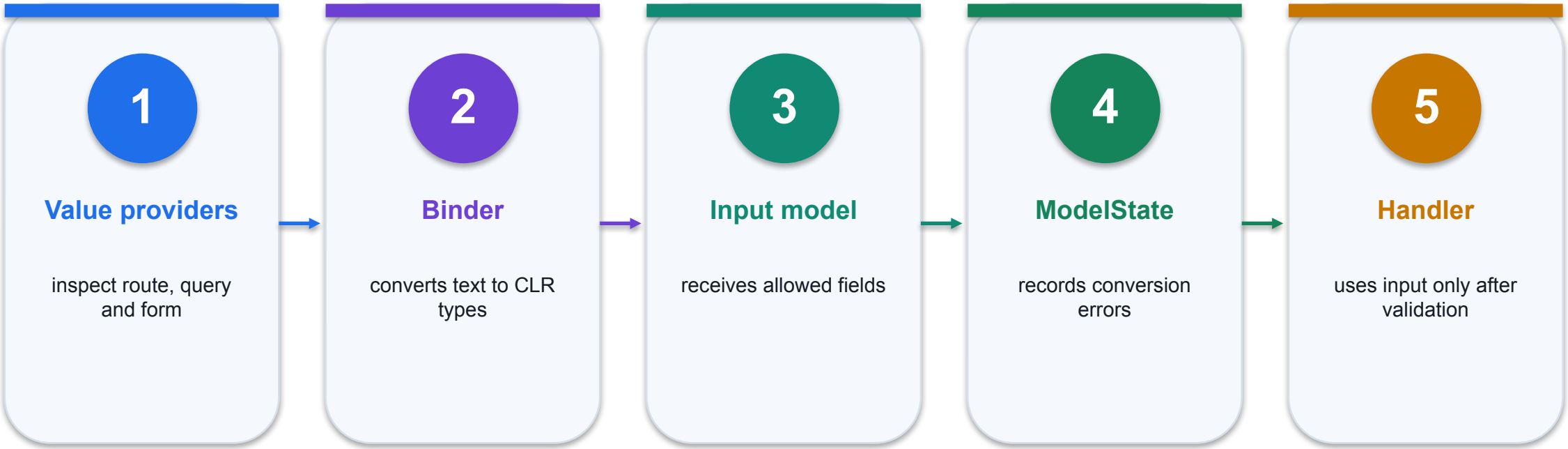
ACCEPTANCE EVIDENCE

- ✓ PageModel unit test substitutes IContactStore
- ✓ Cancellation reaches EF Core
- ✓ Concurrent requests do not share mutable list

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Model binding input sources



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Model binding input sources · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [BindProperty] public NewContact Input { get; set; } = new();
2 public sealed record NewContact(string Summary, string
  Category);
```

1 **[BindProperty]**
binds form property for non-GET by default

2 **SupportsGet**
explicitly allows GET property binding

3 **TryParse**
controls custom conversion

4 **Prefix**
matches nested field names

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Model binding input sources · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
[BindProperty]	binds form property for non-GET by default	Where is this enforced?
SupportsGet	explicitly allows GET property binding	What is observable?
TryParse	controls custom conversion	Who owns the value?
Prefix	matches nested field names	What rejects invalid input?
Overposting	prevented by dedicated input model	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Model binding input sources · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Default values	conversion error was ignored
Overposting	entity was bound directly
Wrong prefix	form name does not match property path
GET mutation	SupportsGet was misused for a command

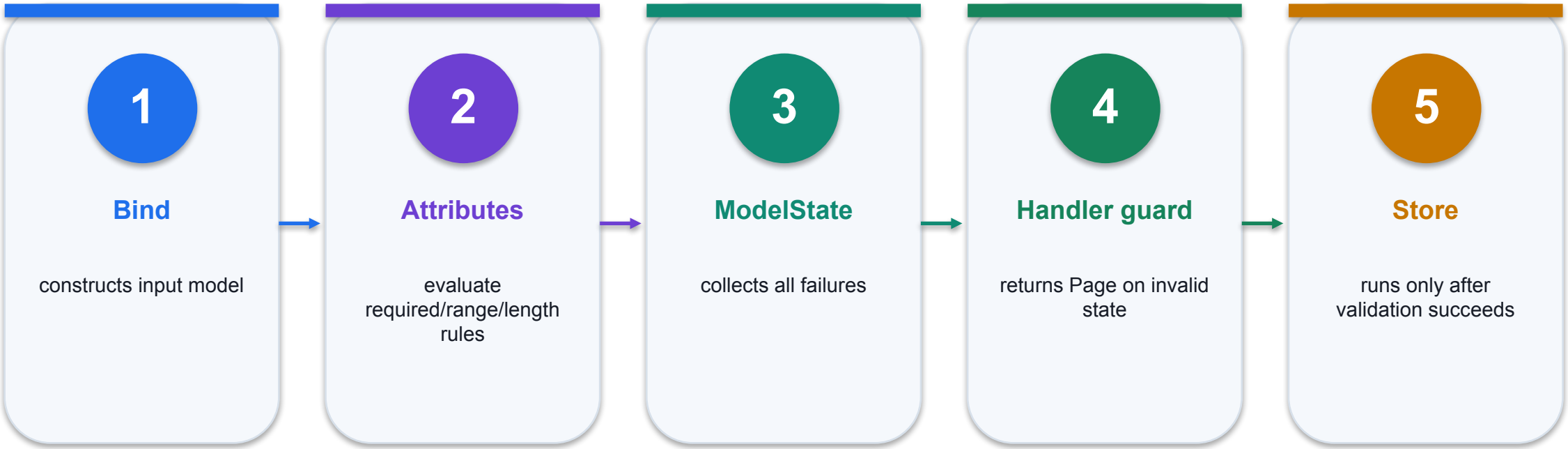
ACCEPTANCE EVIDENCE

- ✓ Invalid integer creates ModelState error
- ✓ Unexpected form field is ignored
- ✓ Valid form populates only NewContact fields

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DataAnnotations and server validation



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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DataAnnotations and server validation · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [Required, EmailAddress]
2 public string Email { get; init; } = string.Empty;
```

1. **[Required]** rejects missing/empty value
2. **[StringLength]** sets bounded text contract
3. **[Range]** sets numeric limits
4. **IValidatableObject** supports cross-field rule

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DataAnnotations and server validation · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
[Required]	rejects missing/empty value	Where is this enforced?
[StringLength]	sets bounded text contract	What is observable?
[Range]	sets numeric limits	Who owns the value?
IValidatableObject	supports cross-field rule	What rejects invalid input?
Database	still enforces storage constraints	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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DataAnnotations and server validation · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Client bypass	server guard was omitted
Mismatched length	UI and database limits differ
Opaque message	default error lacks recovery guidance
Duplicate rule	business validation diverges across endpoints

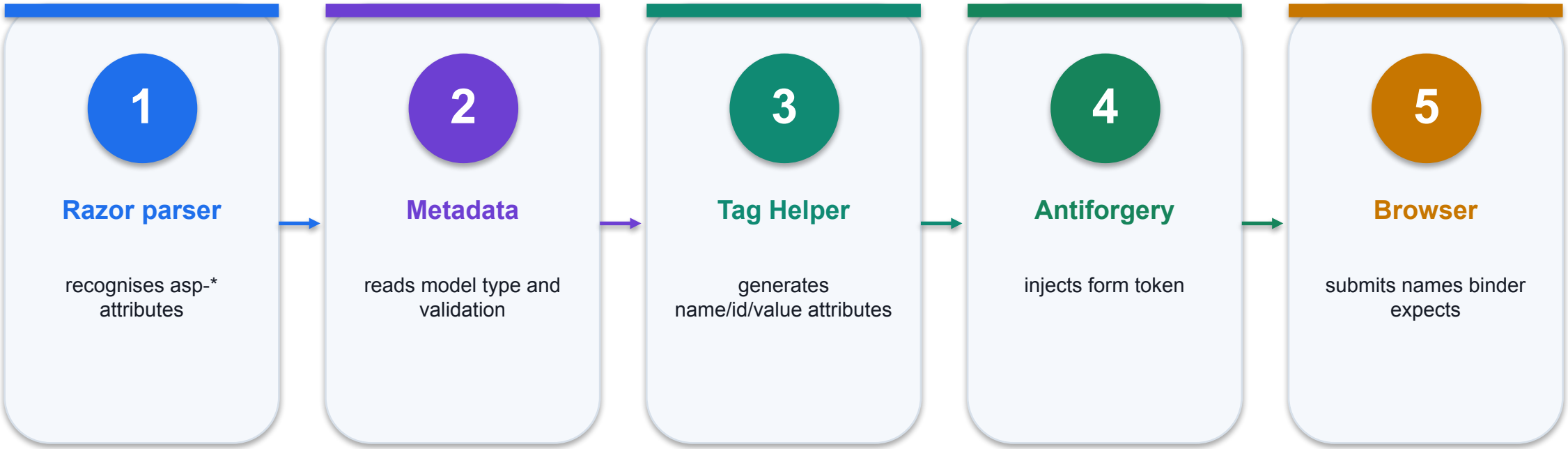
ACCEPTANCE EVIDENCE

- ✓ 4-character summary is rejected
- ✓ 120-character summary succeeds
- ✓ Store mock receives no call for invalid input

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Tag Helpers generate consistent controls



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Tag Helpers generate consistent controls · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <label asp-for="Input.Email"></label>
2 <input asp-for="Input.Email" />
3 <span asp-validation-for="Input.Email"></span>
```

1. **asp-for**
   binds label/input/message to one expression
2. **asp-page**
   generates route-aware URL
3. **asp-validation-for**
   renders ModelState error
4. **form method=post**
   participates in antiforgery

K3 / A3

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Tag Helpers generate consistent controls · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
asp-for	binds label/input/message to one expression	Where is this enforced?
asp-page	generates route-aware URL	What is observable?
asp-validation-for	renders ModelState error	Who owns the value?
form method=post	participates in antiforgery	What rejects invalid input?
partial	reuses reviewed markup	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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TOPIC 3 · DIAGNOSE + VERIFY

Tag Helpers generate consistent controls · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Manual name drift	binder receives no value
Duplicate id	labels and errors bind incorrectly
Raw URL	route refactor breaks link
Missing partial	validation scripts never load

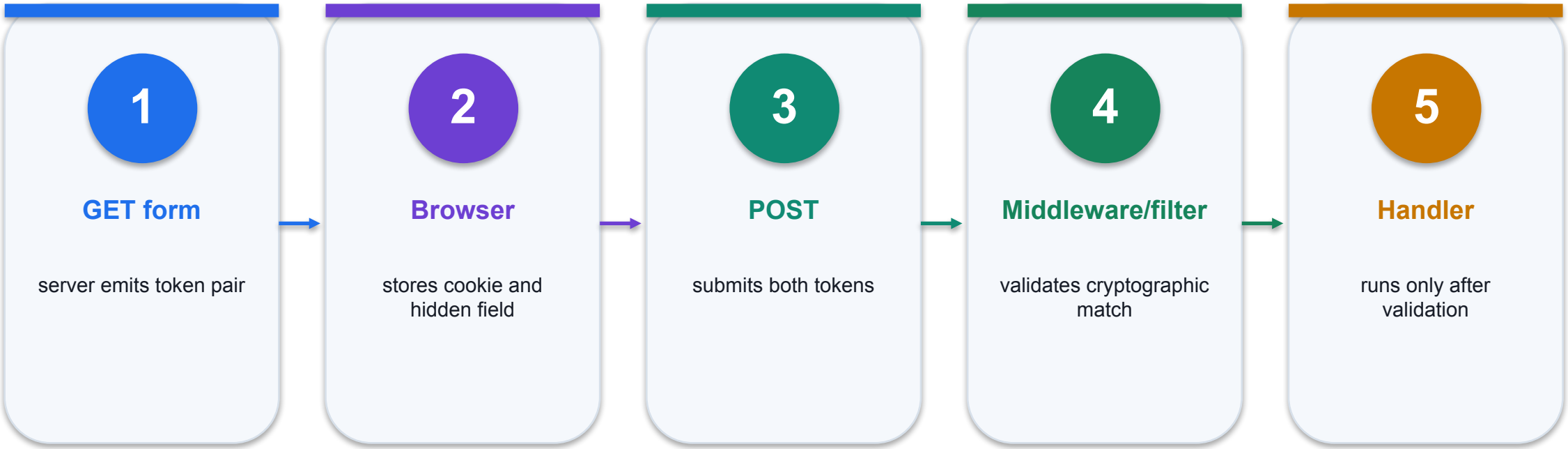
ACCEPTANCE EVIDENCE

- ✓ Rendered label for equals input id
- ✓ Invalid POST redisplays message
- ✓ Generated form contains antiforgery token

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Antiforgery request validation



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Antiforgery request validation · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 <form method="post"> <!-- FormTagHelper emits token -->
2   <button type="submit">Create</button>
3 </form>
```

1 **Cookie**
bound to browser session context

2 **Request token**
hidden form value/header

3 **SameSite**
reduces cross-site cookie sending

4 **POST/PUT/DELETE**
unsafe methods require protection

K3 / A3

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Antiforgery request validation · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Cookie	bound to browser session context	Where is this enforced?
Request token	hidden form value/header	What is observable?
SameSite	reduces cross-site cookie sending	Who owns the value?
POST/PUT/DELETE	unsafe methods require protection	What rejects invalid input?
API auth	bearer-token APIs use different CSRF analysis	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Antiforgery request validation · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

400	token pair is absent or invalid
Token disabled	cross-site form can execute command
Mixed auth	cookie API assumed bearer semantics
Cached form	stale token/cookie pair no longer matches

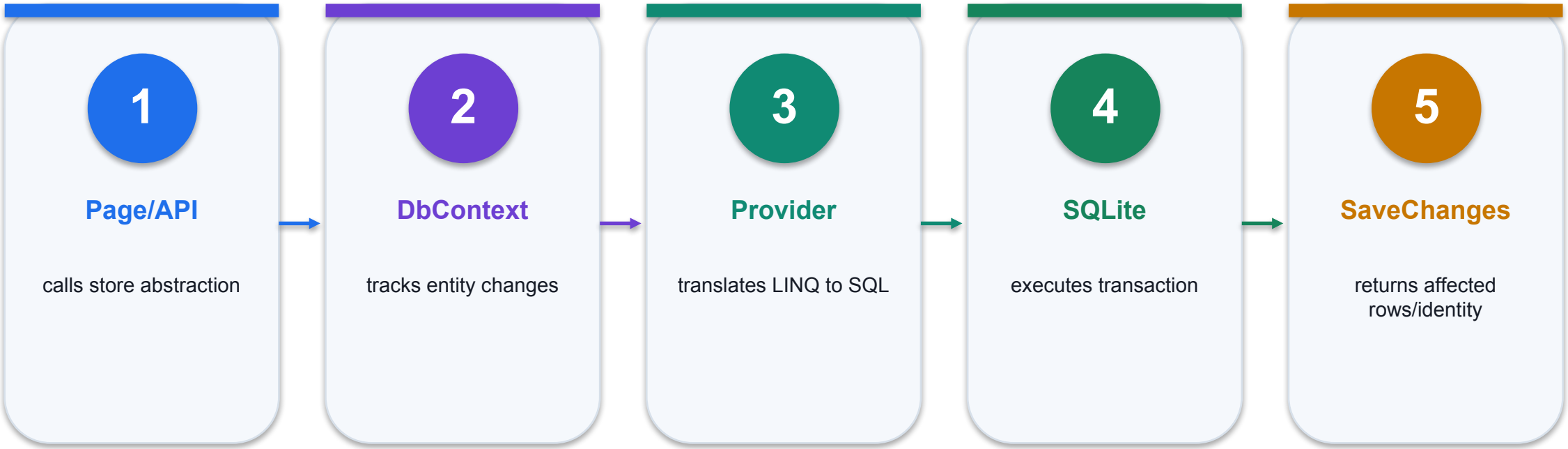
ACCEPTANCE EVIDENCE

- ✓ POST without token returns 400
- ✓ Normal browser POST succeeds
- ✓ Unsafe cookie-auth endpoints are inventoried

Record URL · command · status · build SHA

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EF Core SQLite persistence



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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EF Core SQLite persistence · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  builder.Services.AddDbContext<CrmDbContext>(o =>
2
   o.UseSqlite(builder.Configuration.GetConnectionString("LocalC
   RM")));
```

1 **DbSet**
query and mutation entry point

2 **Migration**
versioned schema change

3 **AsNoTracking**
read-only query optimisation

4 **Async**
does not block while I/O waits

K3 / A3

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EF Core SQLite persistence · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
DbSet	query and mutation entry point	Where is this enforced?
Migration	versioned schema change	What is observable?
AsNoTracking	read-only query optimisation	Who owns the value?
Async	does not block while I/O waits	What rejects invalid input?
Constraint	database protects required/unique rules	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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EF Core SQLite persistence · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

No database	migration was never applied
Locked	long SQLite transaction blocks writer
N+1	navigation query issues repeated SQL
Schema drift	manual DB change bypassed migration

ACCEPTANCE EVIDENCE

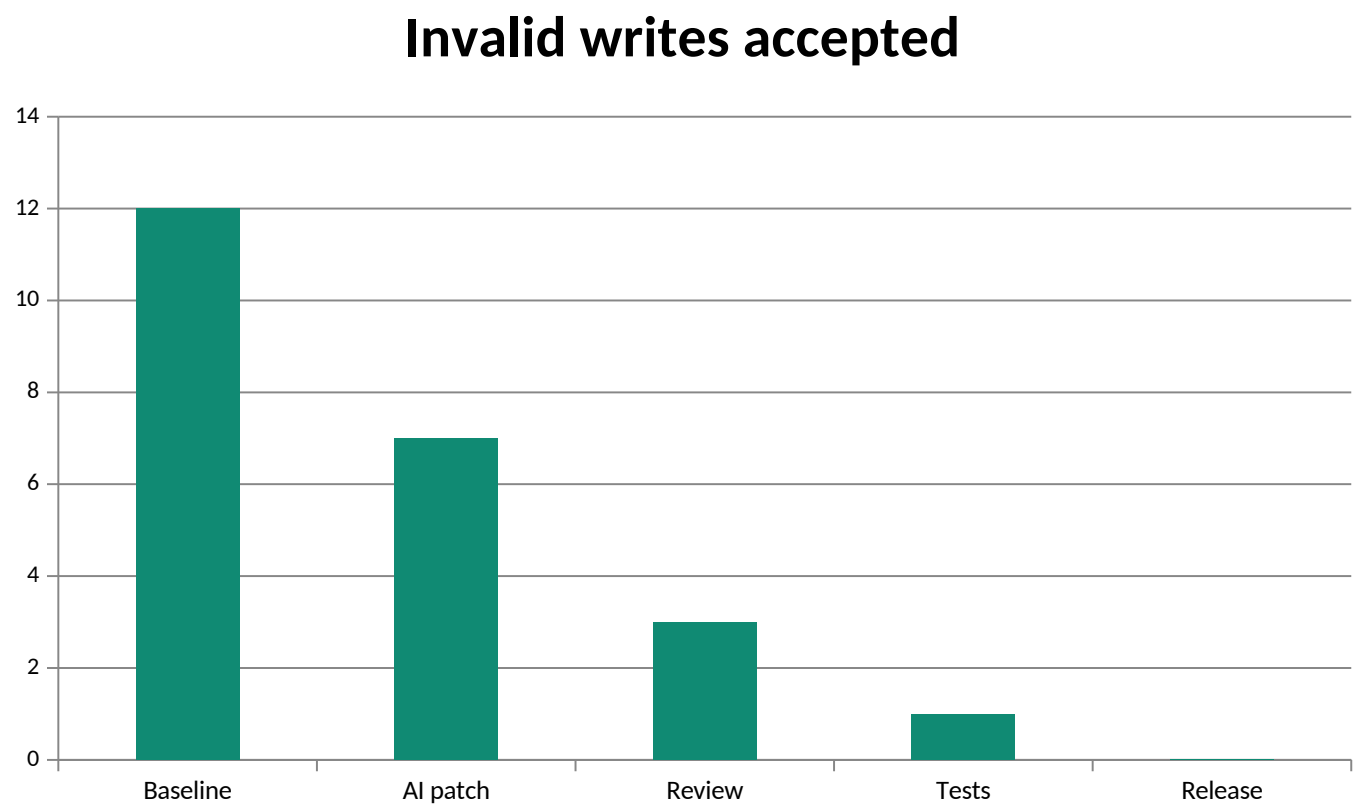
- ✓ dotnet ef database update succeeds
- ✓ Create survives application restart
- ✓ Integration test uses an isolated database

Record URL · command · status · build SHA

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TOPIC 3 · EDITABLE NATIVE CHART

LO3 evidence profile



WHAT THE DATA SHOWS

Server validation and antiforgery close unsafe input paths.

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

EDITABLE CHART

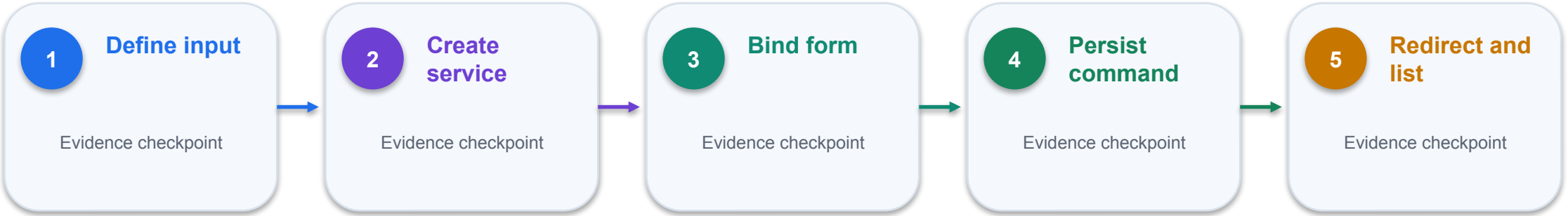
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TOPIC 3 · HANDS-ON ACTIVITY · LO3

65 MIN

Activity 5: Create and List CRM Contacts with Razor Pages

Implement page-focused create/list handlers through an injectable contact store.



YOU'LL PRODUCE **input model, IContactStore boundary, create/list PageModels and successful PRG navigation**

Local snapshot: activities/activity-05-razor-create-list/src/LocalCRM · full commands in the Learner Guide

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Activity 5 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Implement REQ-001 and REQ-002 with Razor Pages. Introduce IContactStore and an in-memory implementation registered as singleton for this temporary Activity only. Use a dedicated NewContact input model and Post/Redirect/Get. Limit changes to Models, Services, Pages/Contacts and Program.cs.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 5 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: input model, IContactStore boundary, create/list PageModels and successful PRG navigation.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

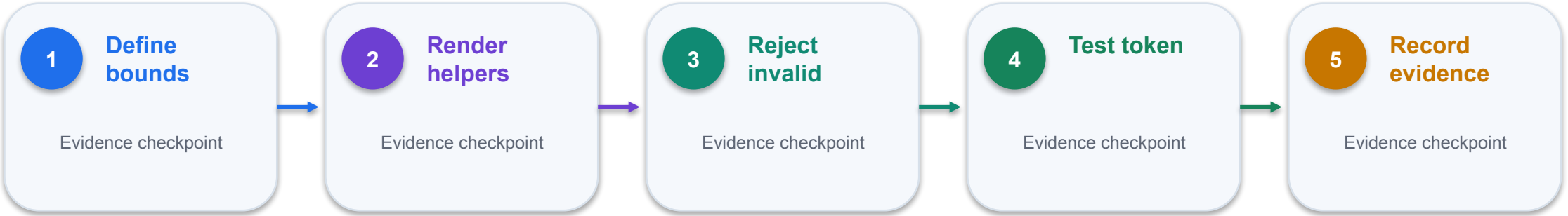
K3/A3 EVIDENCE

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50 MIN

Activity 6: Enforce Validation and Antiforgery Controls

Reject invalid or forged commands while returning accessible recovery guidance.



YOU'LL PRODUCE **bounded DataAnnotations, field/summary errors, antiforgery proof and negative-path tests**

Local snapshot: activities/activity-06-validation-and-antiforgery/src/LocalCRM · full commands in the Learner Guide

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Activity 6 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Harden the Create Razor Page. Name is required (2–80 characters), Email is required and valid, and Stage is Lead, Qualified or Customer. Add accessible field errors and validation summary. Preserve automatic antiforgery and add tests for invalid input and missing token.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 6 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: bounded DataAnnotations, field/summary errors, antiforgery proof and negative-path tests.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

K3/A3 EVIDENCE

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LO3 · K3 / A3

Topic 3 · Evidence Recap

1. **Architecture**
   Name the component boundary and owner.

2. **Mechanism**
   Trace the runtime/data flow in order.

3. **Contract**
   Point to fields, configuration, code or status semantics.

4. **Failure**
   Recognise the named signal and likely cause.

5. **Verification**
   Produce repeatable command or UI evidence.

6. **Activity link**
   Use the cumulative LocalCRM snapshot for this LO.

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TOPIC 4 · LO4 · K4 / A4

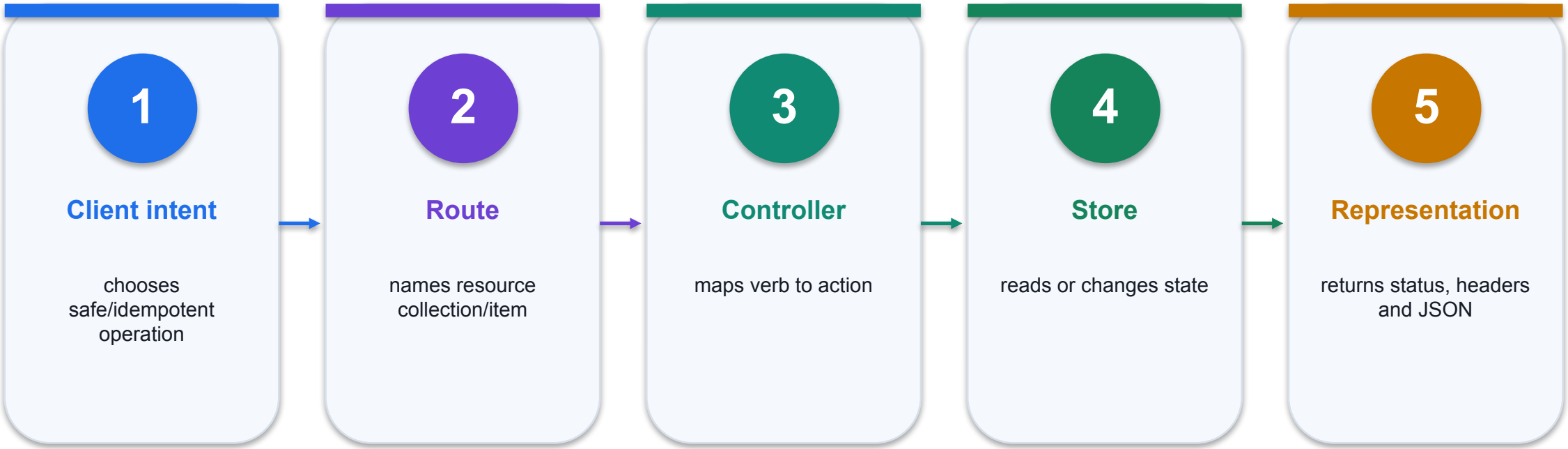
04

Controller APIs and Component Interoperability

resource contract → controller boundary → persistence → observable HTTP evidence

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Resource-oriented HTTP semantics



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Resource-oriented HTTP semantics · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [ApiController]
2 [Route("api/contacts")]
3 public sealed class ContactsController : ControllerBase { }
```

1. **GET /api/contacts**
   list; safe and idempotent
2. **POST /api/contacts**
   create; not inherently idempotent
3. **GET /api/contacts/{id}**
   read one
4. **PUT /api/contacts/{id}**
   replace; idempotent

K4 / A4

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Resource-oriented HTTP semantics · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
GET /api/contacts	list; safe and idempotent	Where is this enforced?
POST /api/contacts	create; not inherently idempotent	What is observable?
GET /api/contacts/{id}	read one	Who owns the value?
PUT /api/contacts/{id}	replace; idempotent	What rejects invalid input?
DELETE /api/contacts/{id}	remove; idempotent outcome	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Resource-oriented HTTP semantics · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Verb tunnel	all commands use POST without semantics
Noun mismatch	route describes implementation, not resource
GET mutates	caches/retries can change state
No Location	client cannot find created resource

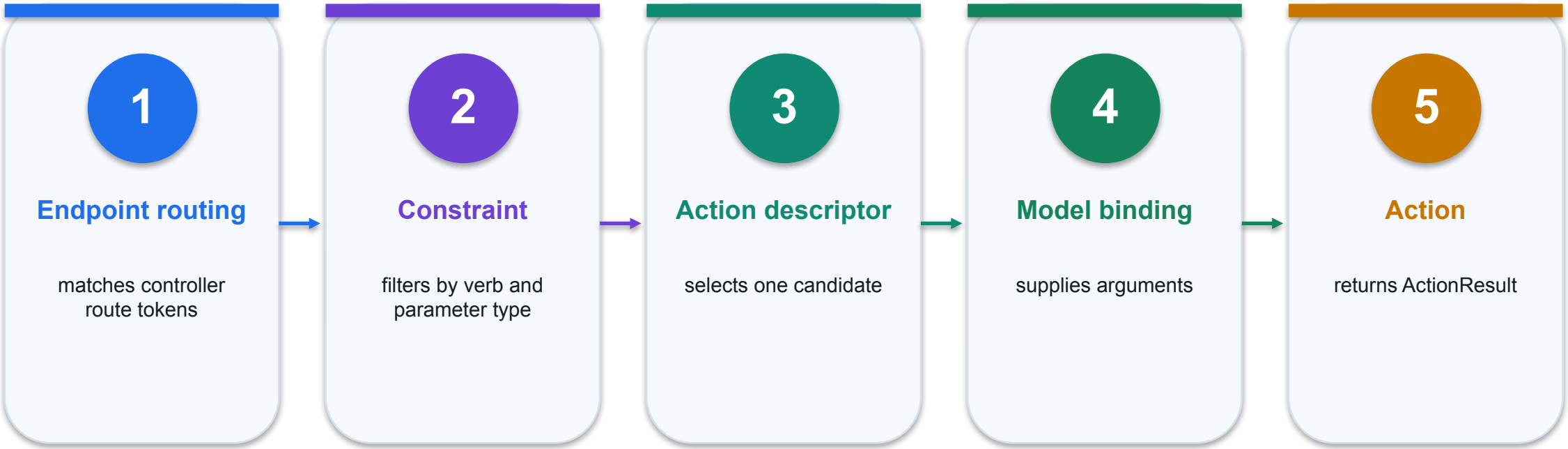
ACCEPTANCE EVIDENCE

- ✓ OpenAPI lists five intended operations
- ✓ GET never changes row count
- ✓ POST returns 201 and Location header

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Controller routing and action selection



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Controller routing and action selection · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 [HttpGet("{id:int}", Name = "GetContact")]
2 public async Task<ActionResult<ContactDto>> Get(int id) {
  /*...*/ }
```

1 **[Route]**
controller-level path prefix

2 **[HttpGet("{id:int}")]**
verb plus constrained template

3 **[FromRoute]**
explicit path source

4 **CreatedAtAction**
generates canonical item URL

K4 / A4

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Controller routing and action selection · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
[Route]	controller-level path prefix	Where is this enforced?
[HttpGet("{id:int}")]	verb plus constrained template	What is observable?
[FromRoute]	explicit path source	Who owns the value?
CreatedAtAction	generates canonical item URL	What rejects invalid input?
Name	supports stable link generation	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Controller routing and action selection · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

AmbiguousMatch	two actions share equivalent route
404	constraint rejects supplied id
Wrong URL	CreatedAtAction route values do not match
Leaked action	public method lacks NonAction/verb attribute

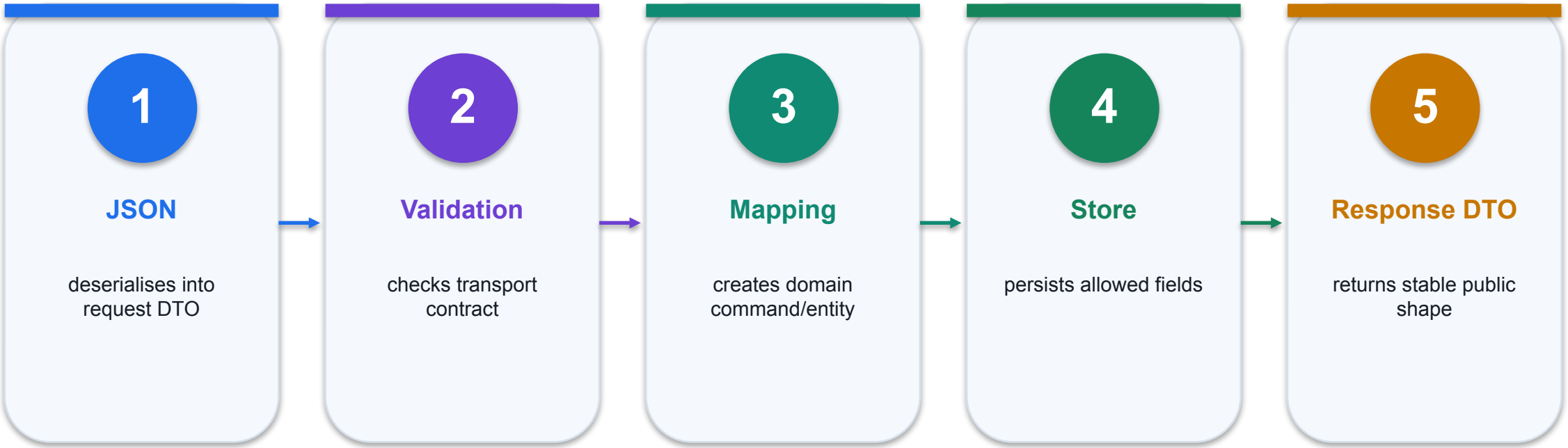
ACCEPTANCE EVIDENCE

- ✓ Route table has no duplicate method+path
- ✓ id=abc produces 404
- ✓ Location URL resolves to the new representation

Record URL · command · status · build SHA

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DTO and domain boundary



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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DTO and domain boundary · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 public sealed record CreateContactDto(
2   [Required, StringLength(80)] string Name,
3   [Required, EmailAddress] string Email, string Stage);
```

1 **CreateContactDto**
client-settable fields only

2 **ContactDto**
id/status/timestamps safe to expose

3 **Entity**
persistence and domain state

4 **Mapping**
explicit code makes changes reviewable

K4 / A4

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DTO and domain boundary · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
CreateContactDto	client-settable fields only	Where is this enforced?
ContactDto	id/status/timestamps safe to expose	What is observable?
Entity	persistence and domain state	Who owns the value?
Mapping	explicit code makes changes reviewable	What rejects invalid input?
Versioning	contract change is deliberate	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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DTO and domain boundary · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Mass assignment	entity is accepted directly
Circular JSON	navigation properties leak
Contract drift	database rename breaks clients
Sensitive field	internal note appears in response

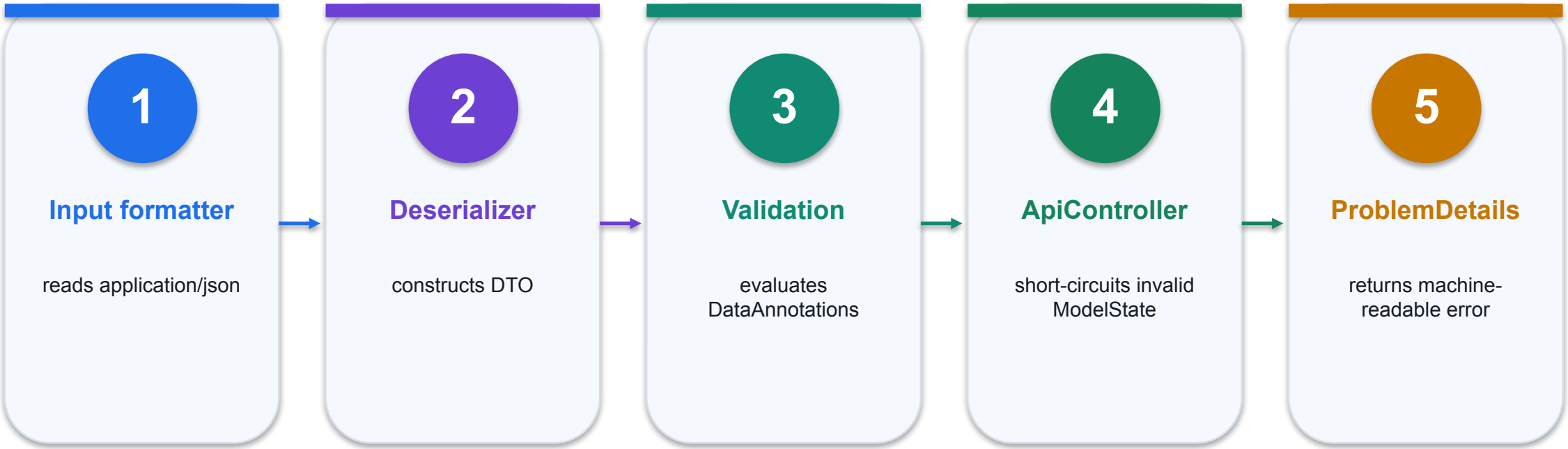
ACCEPTANCE EVIDENCE

- ✓ Unknown JSON property is handled by policy
- ✓ Response excludes internal fields
- ✓ Mapping test covers every public property

Record URL · command · status · build SHA

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API model binding and automatic 400



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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API model binding and automatic 400 · worked artifact

REVIEWED EXCERPT · .NET 10

```
curl -i -X POST /api/contacts \
+ -H "Content-Type: application/json" -d "{\"summary\":\"\"}"
```

1. **Content-Type**
   must select JSON input formatter
2. **400**
   syntactically valid but contract-invalid
3. **415**
   unsupported media type
4. **ValidationProblemDetails**
   errors keyed by field

K4 / A4

Full procedures: Learner Guide

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TOPIC 4 · DESIGN DETAIL

API model binding and automatic 400 · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Content-Type	must select JSON input formatter	Where is this enforced?
400	syntactically valid but contract-invalid	What is observable?
415	unsupported media type	Who owns the value?
ValidationProblemDetails	errors keyed by field	What rejects invalid input?
Nullable	requiredness should be explicit	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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TOPIC 4 · DIAGNOSE + VERIFY

API model binding and automatic 400 · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

400 hidden	client ignores ProblemDetails body
415	Content-Type header is missing/wrong
Empty object	nullable/default contract too permissive
Exception	custom binder throws instead of reporting error

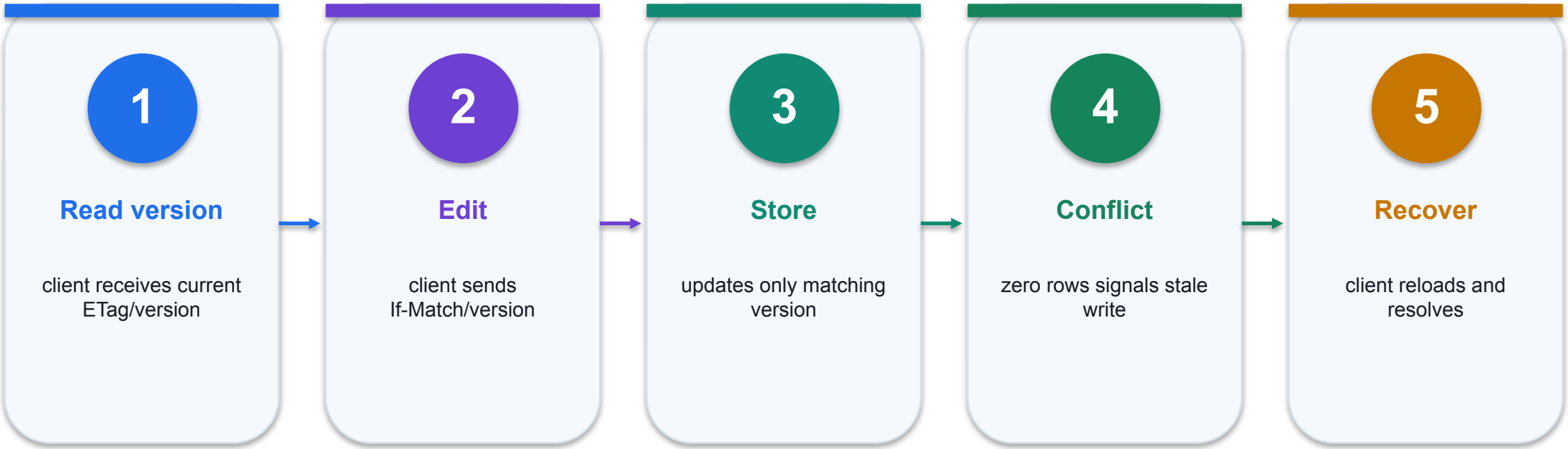
ACCEPTANCE EVIDENCE

- ✓ Invalid DTO returns application/problem+json
- ✓ Error dictionary names summary
- ✓ Controller action is not invoked

Record URL · command · status · build SHA

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CRUD status and concurrency contract



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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CRUD status and concurrency contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 if (dto.Version != entity.Version)
2   return Conflict(new ProblemDetails { Title = "Stale
  contact" });
```

1. **200** — successful read/update with body
2. **201** — created with Location
3. **204** — successful delete without body
4. **404** — resource absent

K4 / A4

Full procedures: Learner Guide

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CRUD status and concurrency contract · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
200	successful read/update with body	Where is this enforced?
201	created with Location	What is observable?
204	successful delete without body	Who owns the value?
404	resource absent	What rejects invalid input?
409/412	state conflict/precondition failed	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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CRUD status and concurrency contract · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Lost update	last writer silently overwrites
Wrong 200	missing resource returns success
Duplicate create	retry creates second record
204 with body	response violates its status semantics

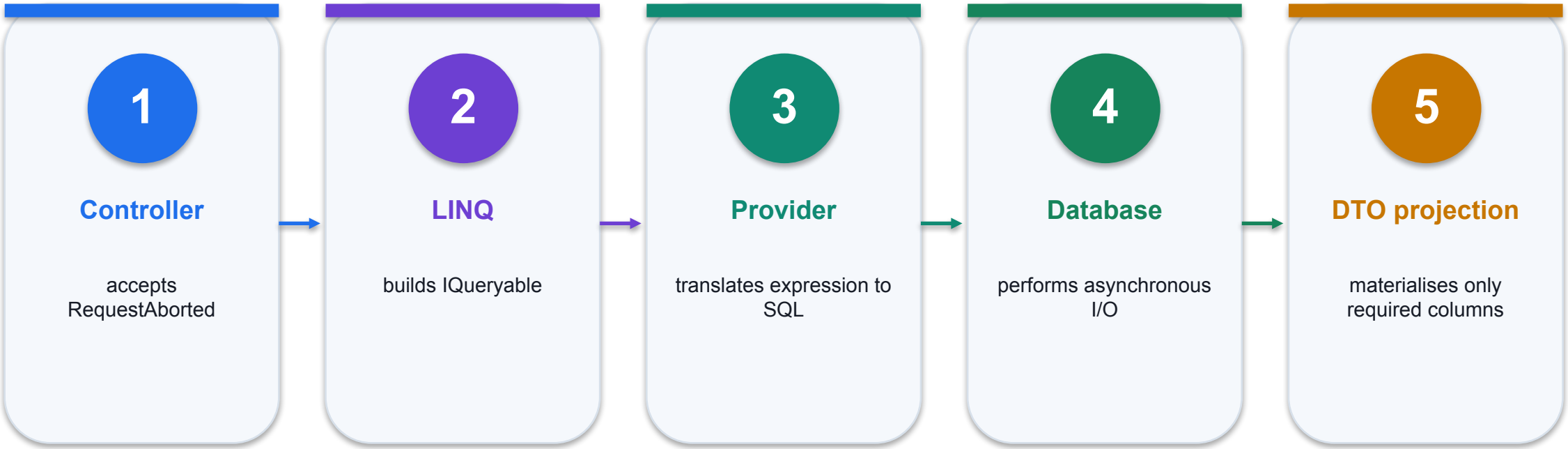
ACCEPTANCE EVIDENCE

- ✓ Stale update produces 409/412
- ✓ Delete missing id produces 404
- ✓ Create provides resolvable Location

Record URL · command · status · build SHA

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Async EF Core API path



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Async EF Core API path · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 var rows = await db.Contacts.AsNoTracking()
2  .Select(x => new ContactDto(x.Id, x.Name, x.Email, x.Stage))
3  .ToListAsync(ct);
```

1 **ToListAsync**
executes query asynchronously

2 **Select**
projects before materialisation

3 **AsNoTracking**
avoids change tracker for reads

4 **CancellationToken**
stops abandoned work

K4 / A4

Full procedures: Learner Guide

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Async EF Core API path · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
ToListAsync	executes query asynchronously	Where is this enforced?
Select	projects before materialisation	What is observable?
AsNoTracking	avoids change tracker for reads	Who owns the value?
CancellationToken	stops abandoned work	What rejects invalid input?
SaveChangesAsync	persists transaction	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Async EF Core API path · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Client eval	expression cannot translate
Over-fetch	entity graph is loaded unnecessarily
No cancellation	work continues after disconnect
Parallel DbContext	same context is used concurrently

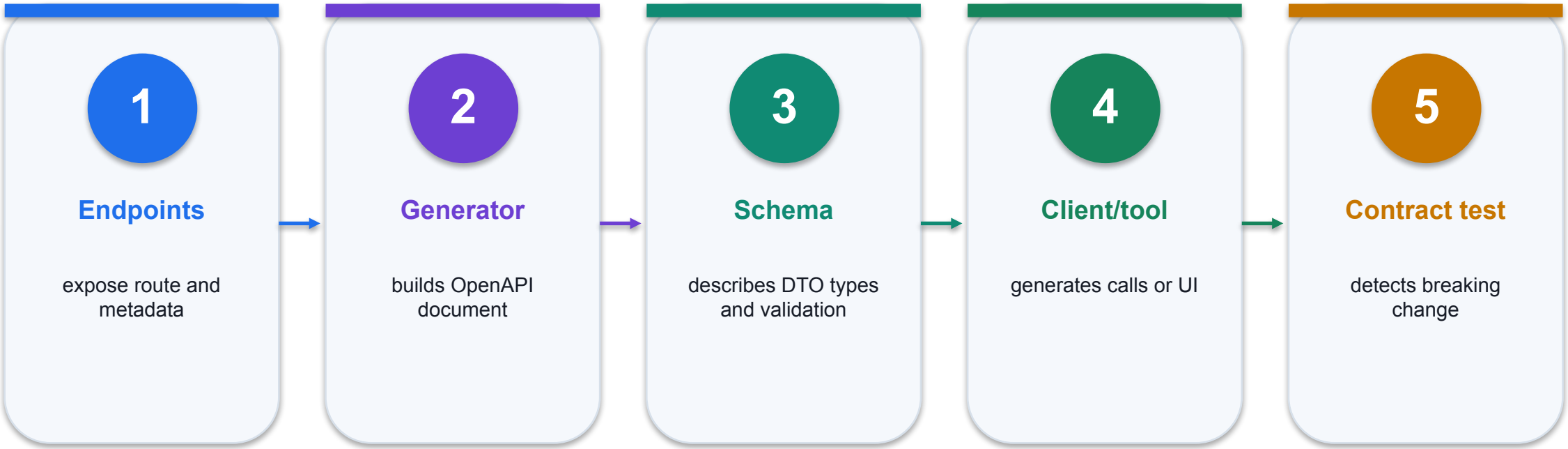
ACCEPTANCE EVIDENCE

- ✓ SQL log selects only DTO columns
- ✓ Cancelled request ends query
- ✓ Load test shows no sync blocking hotspot

Record URL · command · status · build SHA

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OpenAPI as interoperability contract



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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OpenAPI as interoperability contract · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 builder.Services.AddOpenApi();
2 if (app.Environment.IsDevelopment()) app.MapOpenApi();
```

1 **paths**
method and route operations

2 **operationId**
stable generated-client method

3 **components/schemas**
reusable JSON shapes

4 **responses**
document every intended status

K4 / A4

Full procedures: Learner Guide

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OpenAPI as interoperability contract · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
paths	method and route operations	Where is this enforced?
operationId	stable generated-client method	What is observable?
components/schemas	reusable JSON shapes	Who owns the value?
responses	document every intended status	What rejects invalid input?
securitySchemes	declares authentication mechanism	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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OpenAPI as interoperability contract · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Undocumented status	client cannot model failure
Schema leak	entity exposes internal fields
Breaking rename	generated client no longer compiles
Prod exposure	interactive docs reveal attack surface

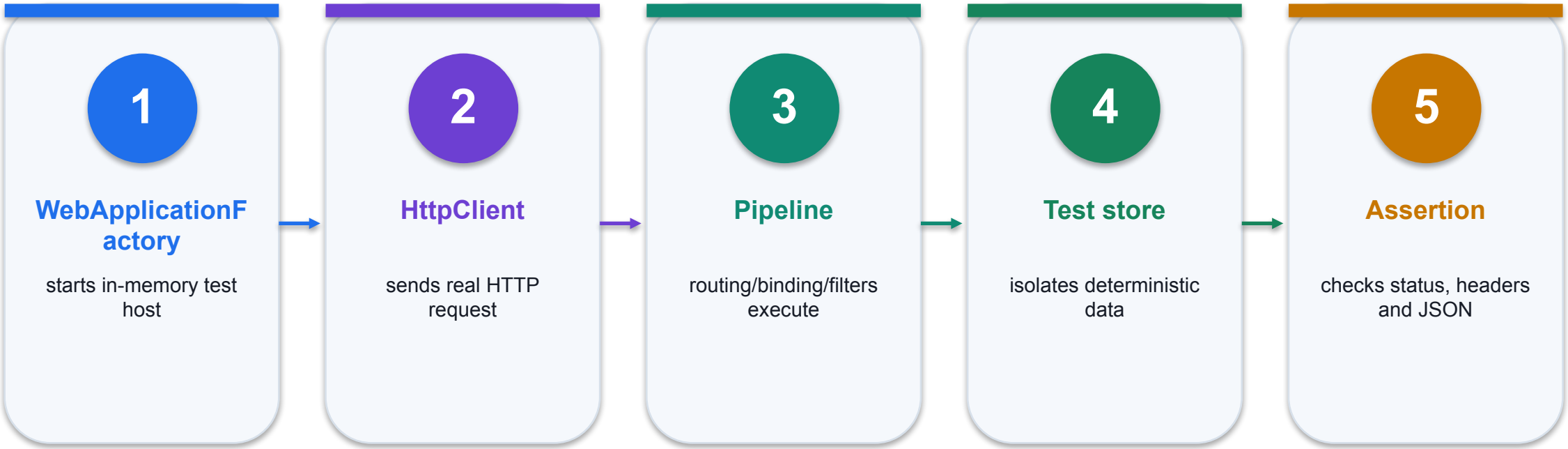
ACCEPTANCE EVIDENCE

- ✓ openapi.json validates
- ✓ Snapshot diff contains no accidental breaking change
- ✓ 201/400/404 schemas are declared

Record URL · command · status · build SHA

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Integration tests and ProblemDetails



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Integration tests and ProblemDetails · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  var response = await client.PostAsJsonAsync("/api/contacts",
   new { summary = "" });
2  Assert.Equal(HttpStatusCode.BadRequest, response.StatusCode);
```

1 **Arrange**
seed isolated database

2 **Act**
call public endpoint only

3 **Assert**
status + content type + shape

4 **ProblemDetails**
stable error envelope

K4 / A4

Full procedures: Learner Guide

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Integration tests and ProblemDetails · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Arrange	seed isolated database	Where is this enforced?
Act	call public endpoint only	What is observable?
Assert	status + content type + shape	Who owns the value?
ProblemDetails	stable error envelope	What rejects invalid input?
Trace id	links client failure to server log	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Integration tests and ProblemDetails · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Unit-only	middleware/routing regression is invisible
Shared DB	tests affect each other
Body-only	wrong status still passes
Stack trace leak	error response exposes internals

ACCEPTANCE EVIDENCE

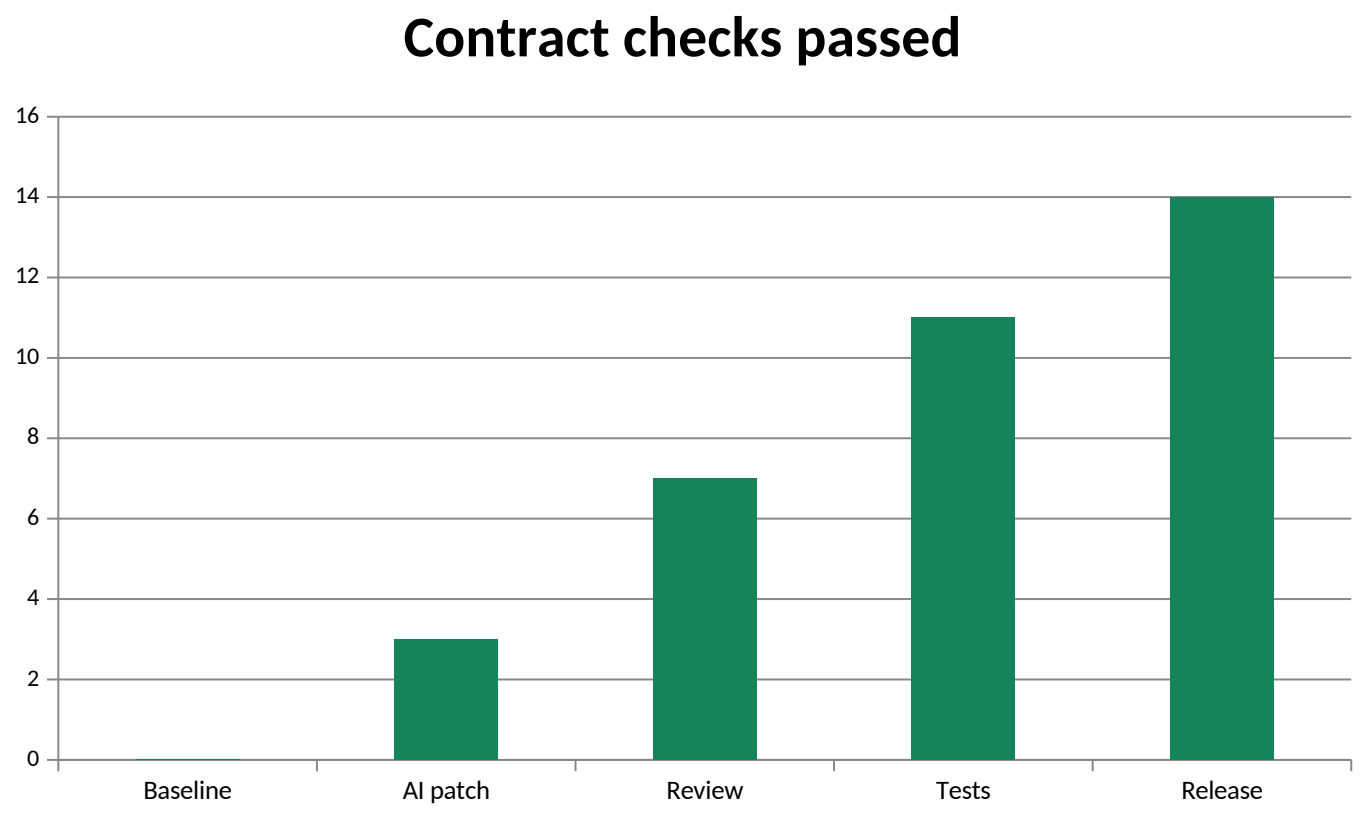
- ✓ Happy and invalid paths run through HTTP
- ✓ ProblemDetails contains traceId
- ✓ Tests pass in parallel with isolated state

Record URL · command · status · build SHA

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TOPIC 4 · EDITABLE NATIVE CHART

LO4 evidence profile



WHAT THE DATA SHOWS

Interoperability confidence comes from status, header and schema assertions.

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

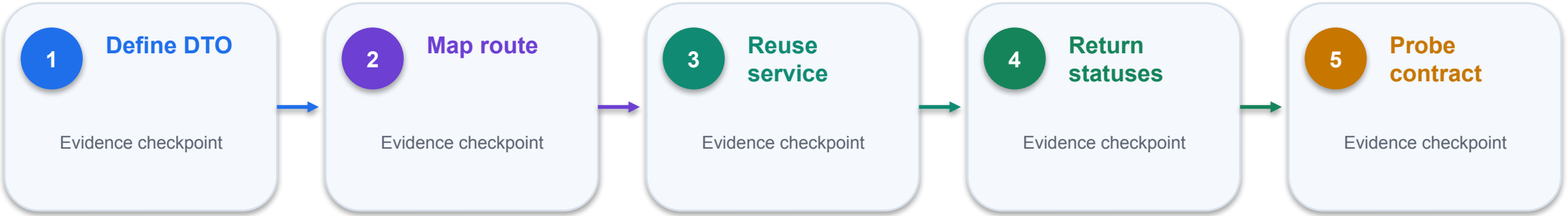
EDITABLE CHART

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65 MIN

Activity 7: Expose a Controller-Based CRUD API

Add a JSON controller surface that reuses the application service without exposing persistence entities.



YOU'LL PRODUCE **controller routes, DTOs, 201/200/400/404 behaviour and curl evidence**

Local snapshot: activities/activity-07-controller-api/src/LocalCRM · full commands in the Learner Guide

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Activity 7 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Add an ApiController at /api/contacts that supports GET collection, GET item and POST create. Use transport DTOs, IContactStore, CreatedAtAction and CancellationToken. Do not return persistence entities directly. Include 400 and 404 evidence.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 7 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: controller routes, DTOs, 201/200/400/404 behaviour and curl evidence.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

K4/A4 EVIDENCE

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70 MIN

Activity 8: Prove Persistence and Interoperability

Replace volatile state with EF Core SQLite, publish an OpenAPI contract and verify components through integration tests.



YOU'LL PRODUCE **SQLite migration, persistent CRUD path, openapi.json and HTTP integration tests**

Local snapshot: activities/activity-08-sqlite-openapi-tests/src/LocalCRM · full commands in the Learner Guide

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Activity 8 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Replace InMemoryContactStore with EF Core SQLite behind IContactStore. Add a migration, async cancellation-aware queries, OpenAPI in Development, and WebApplicationFactory tests for 201, 400 and 404. Preserve existing page and API contracts.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 8 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: SQLite migration, persistent CRUD path, openapi.json and HTTP integration tests.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

K4/A4 EVIDENCE

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LO4 · K4 / A4

Topic 4 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.

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TOPIC 5 · LO5 · K5 / A5

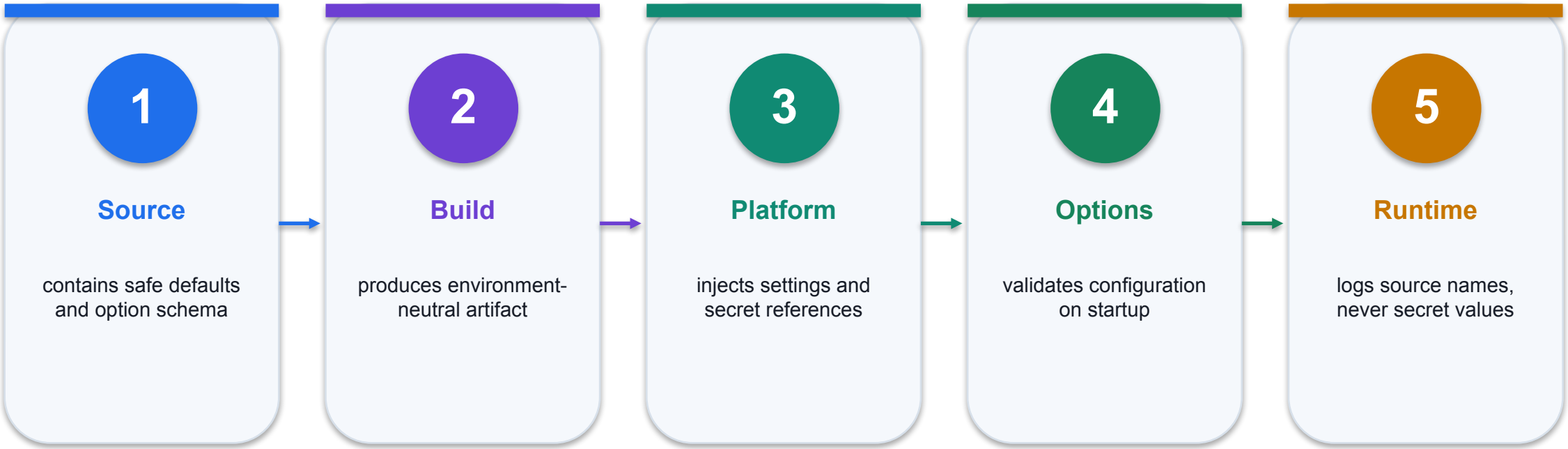
05

Production Hosting, Azure and Design Evidence

deployable artifact → observable service → safe release → traceable design record

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Environment-specific configuration and secrets



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Environment-specific configuration and secrets · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  var cs =
   builder.Configuration.GetConnectionString("LocalCRM")
2    ?? throw new InvalidOperationException("Missing LocalCRM
   connection");
```

1 **ASPNETCORE_ENVIRONMENT**
Development/Staging/Production

2 **App__FeatureFlag**
environment variable key encoding

3 **Key Vault reference**
secret stays outside app settings

4 **ValidateOnStart**
fails before traffic

K5 / A5

Full procedures: Learner Guide

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Environment-specific configuration and secrets · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
ASPNETCORE_ENVIRONMENT	Development/Staging/Production	Where is this enforced?
App__FeatureFlag	environment variable key encoding	What is observable?
Key Vault reference	secret stays outside app settings	Who owns the value?
ValidateOnStart	fails before traffic	What rejects invalid input?
Rotation	changes secret without source edit	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Environment-specific configuration and secrets · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Secret in repo	credential entered appsettings.json
Wrong slot value	setting was not marked deployment-slot specific
Silent default	production uses developer fallback
Log leak	configuration object was serialised

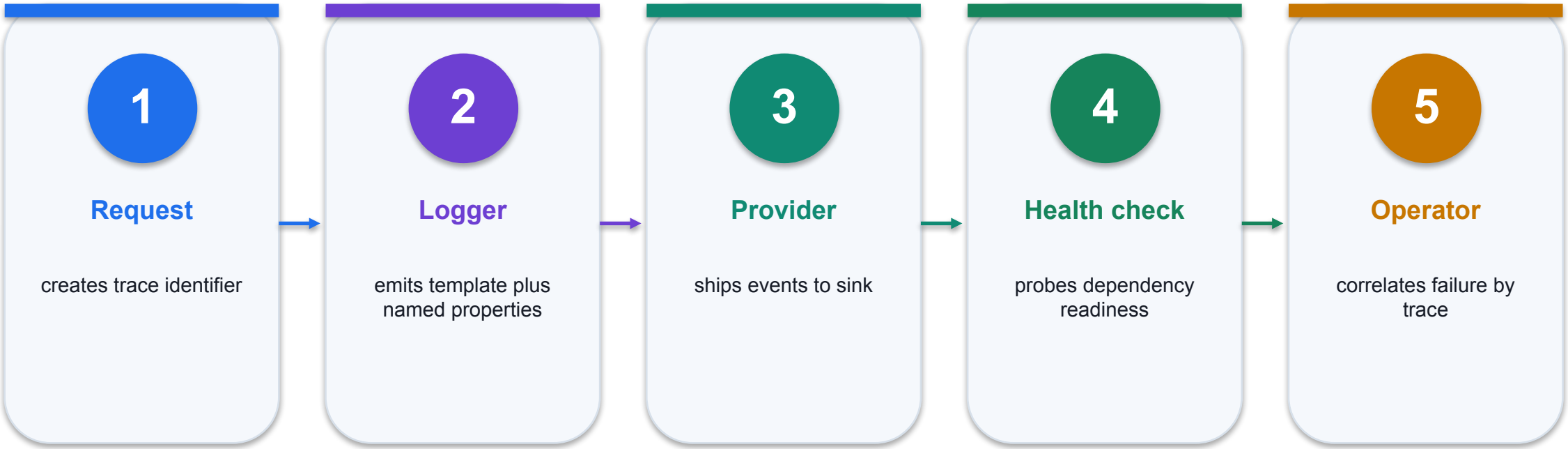
ACCEPTANCE EVIDENCE

- ✓ Secret scan is clean
- ✓ Production start fails without required setting
- ✓ Rotated value is picked up via platform process

Record URL · command · status · build SHA

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Structured logging and health checks



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Structured logging and health checks · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 builder.Services.AddHealthChecks().AddDbContextCheck<CrmDbCon
  text>();
2 app.MapHealthChecks("/health/ready");
```

1 **Template**
Contact {ContactId} created

2 **Level**
Information/Warning/Error chosen by action

3 **Liveness**
process is running

4 **Readiness**
dependencies can serve traffic

K5 / A5

Full procedures: Learner Guide

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Structured logging and health checks · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Template	Contact {ContactId} created	Where is this enforced?
Level	Information/Warning/Error chosen by action	What is observable?
Liveness	process is running	Who owns the value?
Readiness	dependencies can serve traffic	What rejects invalid input?
PII	not emitted as log property	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Structured logging and health checks · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

String log	fields cannot be queried
False green	liveness ignores failed database
PII leak	request body is logged
No trace	client error cannot be correlated

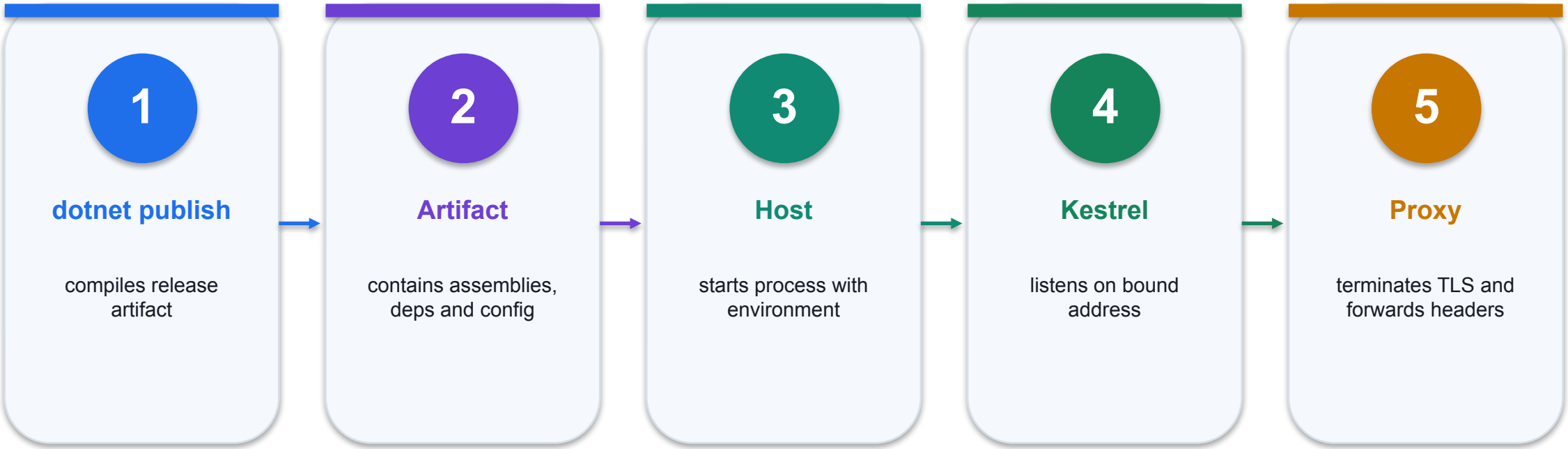
ACCEPTANCE EVIDENCE

- ✓ /health/live and /health/ready have distinct meaning
- ✓ 500 log carries trace id
- ✓ No sensitive payload appears in sink

Record URL · command · status · build SHA

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Publish output and hosting boundary



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Publish output and hosting boundary · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 dotnet publish -c Release -o ./publish
2 ASPNETCORE_URLS=http://+:8080 dotnet ./publish/LocalCRM.dll
```

1. **Framework-dependent**
   small artifact requires installed runtime
2. **Self-contained**
   larger artifact bundles runtime
3. **ForwardedHeaders**
   trust only known proxy
4. **PORT/URLS**
   platform supplies binding

K5 / A5

Full procedures: Learner Guide

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Publish output and hosting boundary · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Framework-dependent	small artifact requires installed runtime	Where is this enforced?
Self-contained	larger artifact bundles runtime	What is observable?
ForwardedHeaders	trust only known proxy	Who owns the value?
PORT/URLS	platform supplies binding	What rejects invalid input?
Shutdown	SIGTERM allows graceful stop	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Publish output and hosting boundary · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

502	process crashed or bound wrong port
Redirect loop	proxy headers are not handled
Runtime missing	framework-dependent target lacks .NET
Partial artifact	publish folder was copied inconsistently

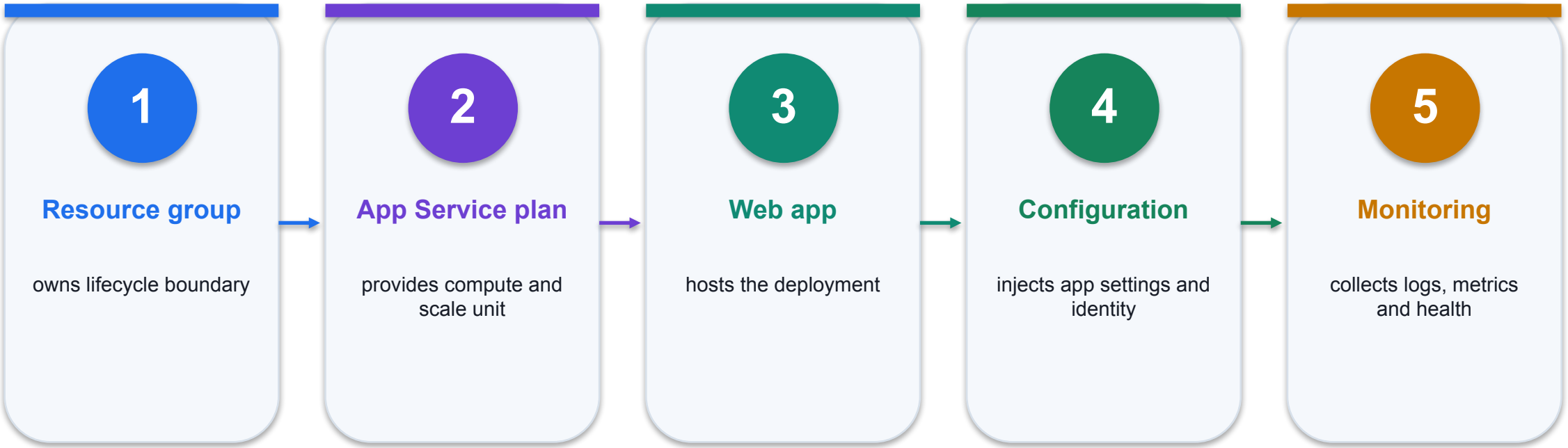
ACCEPTANCE EVIDENCE

- ✓ Release artifact starts in a clean directory
- ✓ HTTP probe reaches bound port
- ✓ SIGTERM stops without dropping active request

Record URL · command · status · build SHA

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Azure App Service component model



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Azure App Service component model · worked artifact

REVIEWED EXCERPT · .NET 10

```
az webapp config appsettings set --resource-group rg-localcrm \
+ --name localcrm-web --settings ASPNETCORE_ENVIRONMENT=Production
```

1. **Region** — latency/data residency decision
2. **Plan SKU** — cost, scale and feature envelope
3. **Linux/Windows** — runtime hosting choice
4. **Always On** — avoids idle unload where supported

K5 / A5

Full procedures: Learner Guide

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Azure App Service component model · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Region	latency/data residency decision	Where is this enforced?
Plan SKU	cost, scale and feature envelope	What is observable?
Linux/Windows	runtime hosting choice	Who owns the value?
Always On	avoids idle unload where supported	What rejects invalid input?
Health check	removes unhealthy instances	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Azure App Service component model · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Wrong plan	required feature is unavailable
Cold/idle	Always On not enabled where needed
Unhealthy instance	health path not configured
Configuration drift	portal-only edits bypass design record

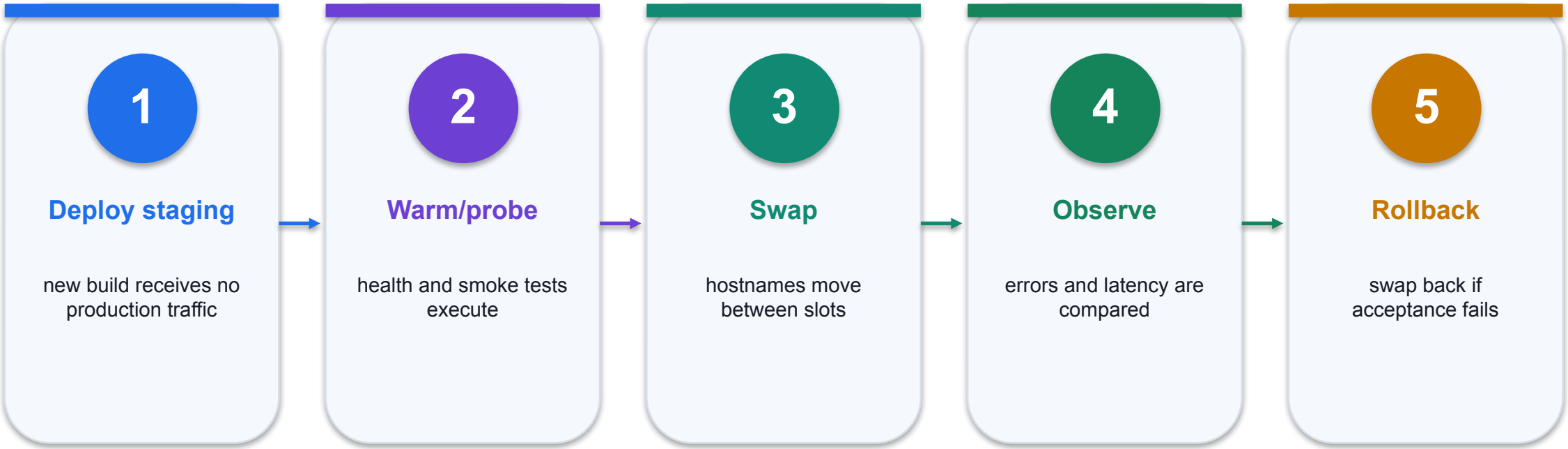
ACCEPTANCE EVIDENCE

- ✓ Resource inventory names RG, plan and app
- ✓ Health check path is configured
- ✓ Deployment settings match documented environment

Record URL · command · status · build SHA

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Deployment slots and safe promotion



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Deployment slots and safe promotion · worked artifact

REVIEWED EXCERPT · .NET 10

```
az webapp deployment slot swap -g rg-localcrm -n localcrm-web \
+ --slot staging --target-slot production
```

1 Slot setting
secret/connection stays with slot

2 Warm-up
application initialises before swap

3 Preview
validates configuration changes

4 Smoke test
uses public contract

K5 / A5

Full procedures: Learner Guide

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Deployment slots and safe promotion · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Slot setting	secret/connection stays with slot	Where is this enforced?
Warm-up	application initialises before swap	What is observable?
Preview	validates configuration changes	Who owns the value?
Smoke test	uses public contract	What rejects invalid input?
Rollback	predefined threshold and command	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Deployment slots and safe promotion · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Bad swap	slot-specific setting was not sticky
Warm-up miss	first production request compiles/initialises
No rollback	failed build remains live
Schema incompatibility	old app cannot run after migration

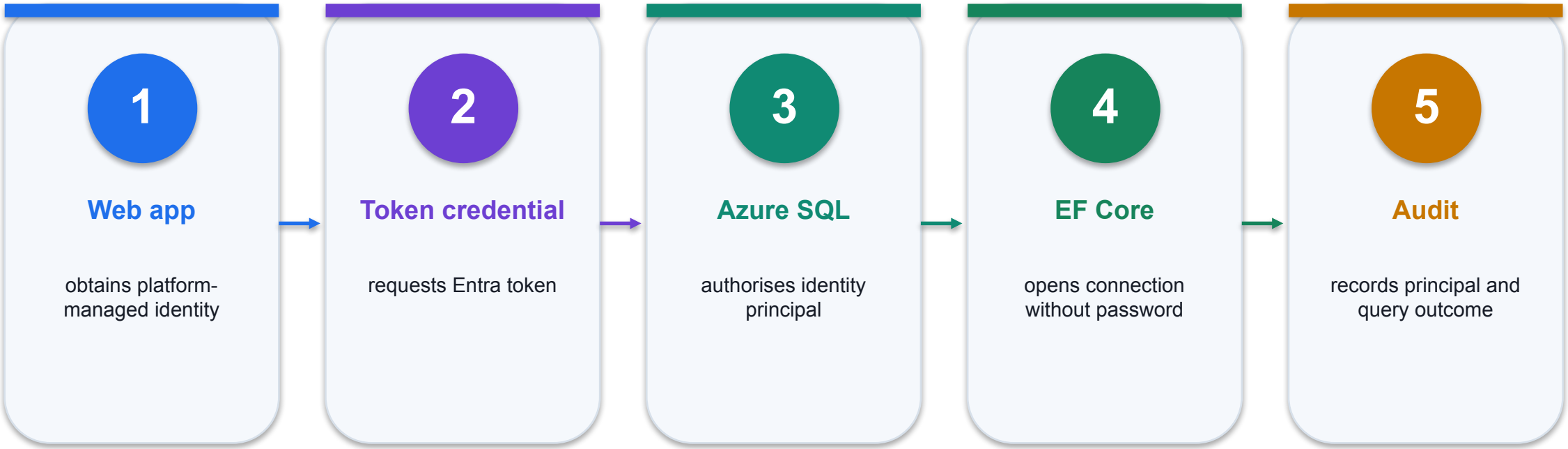
ACCEPTANCE EVIDENCE

- ✓ Staging smoke test passes before swap
- ✓ Production health remains green
- ✓ Rollback step is rehearsed and timed

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Managed identity and database access



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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Managed identity and database access · worked artifact

REVIEWED EXCERPT · .NET 10

```
1  builder.Services.AddAzureClients(c => c.UseCredential(new
   DefaultAzureCredential()));
```

1 **System assigned**
identity lifecycle follows app

2 **User assigned**
identity can be shared/reused

3 **Least privilege**
grant only required database role

4 **DefaultAzureCredential**
local and hosted credential chain

K5 / A5

Full procedures: Learner Guide

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Managed identity and database access · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
System assigned	identity lifecycle follows app	Where is this enforced?
User assigned	identity can be shared/reused	What is observable?
Least privilege	grant only required database role	Who owns the value?
DefaultAzureCredential	local and hosted credential chain	What rejects invalid input?
No password	rotation burden moves to platform	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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Managed identity and database access · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Login failed	identity lacks database user/role
Wrong tenant	credential resolved unexpected account
Overprivilege	app identity is database owner
Local mismatch	developer chain differs from hosted identity

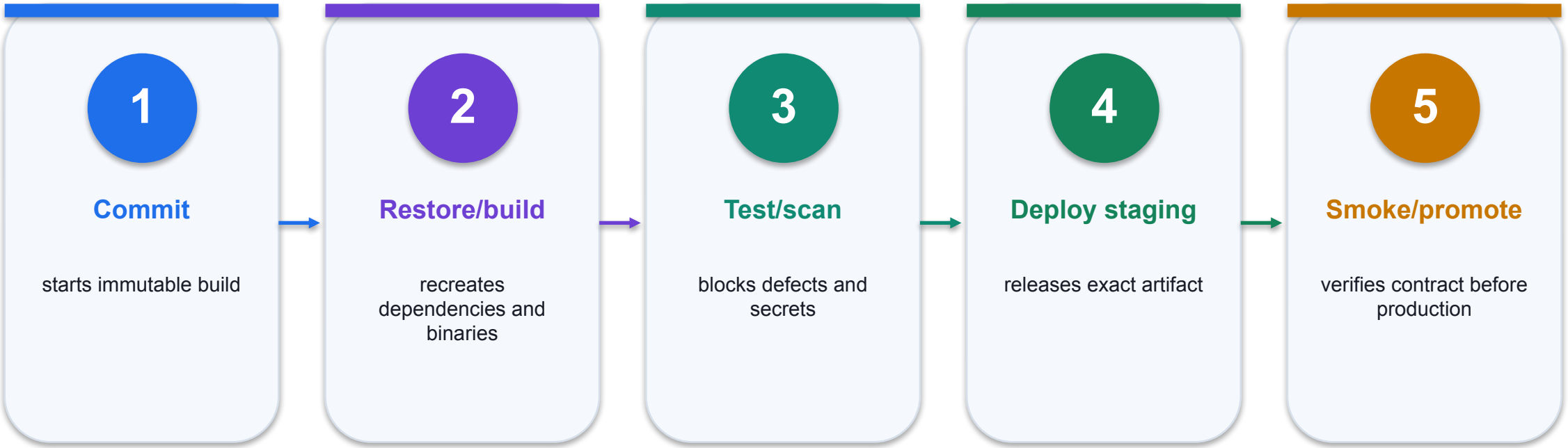
ACCEPTANCE EVIDENCE

- ✓ App connects with passwordless identity
- ✓ Database audit names the app principal
- ✓ Denied operation proves least privilege

Record URL · command · status · build SHA

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CI/CD gates and smoke evidence



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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CI/CD gates and smoke evidence · worked artifact

REVIEWED EXCERPT · .NET 10

```
1 dotnet restore --locked-mode
2 dotnet build -c Release --no-restore
3 dotnet test -c Release --no-build
```

1 **Artifact SHA**
links deployed bytes to commit

2 **dotnet test**
unit and integration gate

3 **Dependency scan**
known-vulnerability control

4 **Environment approval**
separates build from promotion

K5 / A5

Full procedures: Learner Guide

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CI/CD gates and smoke evidence · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Artifact SHA	links deployed bytes to commit	Where is this enforced?
dotnet test	unit and integration gate	What is observable?
Dependency scan	known-vulnerability control	Who owns the value?
Environment approval	separates build from promotion	What rejects invalid input?
Smoke	health plus one business transaction	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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CI/CD gates and smoke evidence · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Mutable deploy	server rebuilds from source
No lock	dependency version changes silently
Green build/bad app	no HTTP smoke test
Untraceable	artifact SHA not recorded

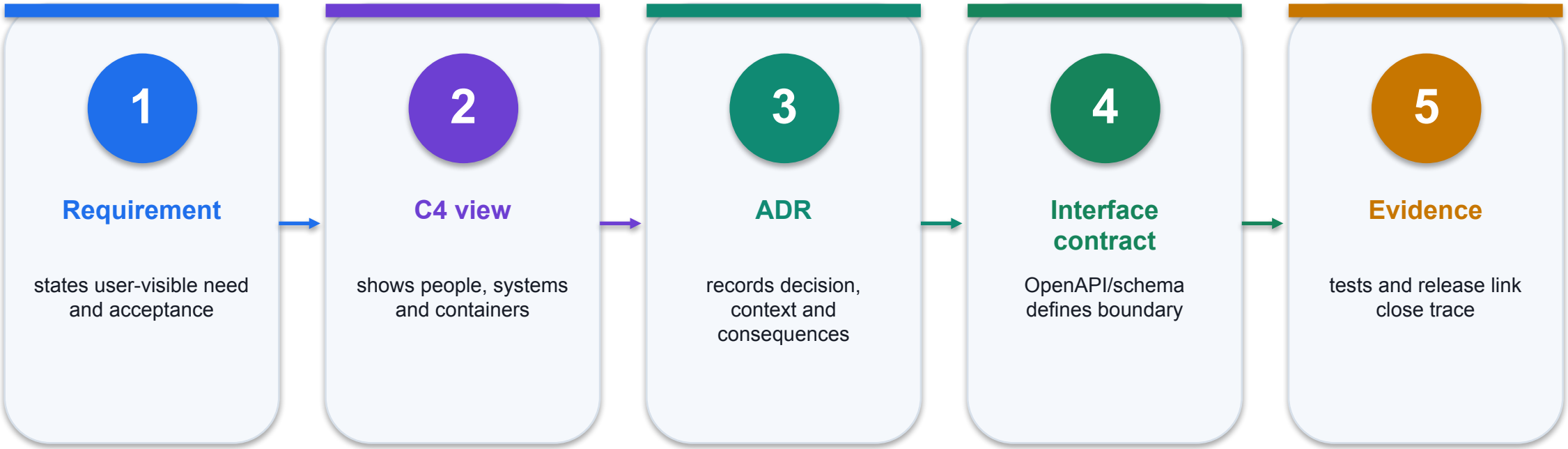
ACCEPTANCE EVIDENCE

- ✓ Pipeline recreates build from lock file
- ✓ Artifact hash is recorded
- ✓ Smoke test proves health and create/read path

Record URL · command · status · build SHA

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ADR, C4 and requirements traceability



DESIGN RULE Trace the boundary end to end; verify both the expected response and the failure that proves the boundary is enforced.

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ADR, C4 and requirements traceability · worked artifact

REVIEWED EXCERPT · .NET 10

```
# ADR-0001: Razor Pages + controller API
Status: Accepted
Decision: page-focused UI; JSON API boundary
Consequences: shared service; two HTTP surfaces.
```

1 **Requirement ID**
REQ-001 stable identifier

2 **Decision**
ADR-0001 accepted/rejected options

3 **Component**
owner and responsibility

4 **Interface**
method/path/schema/status

K5 / A5

Full procedures: Learner Guide

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ADR, C4 and requirements traceability · interface contract

CONTRACT ELEMENT	OPERATIONAL MEANING	REVIEW QUESTION
Requirement ID	REQ-001 stable identifier	Where is this enforced?
Decision	ADR-0001 accepted/rejected options	What is observable?
Component	owner and responsibility	Who owns the value?
Interface	method/path/schema/status	What rejects invalid input?
Verification	test ID, result and build SHA	What must remain stable?

Vibe-coding gate: accept generated code only when every contract row is visible in code, tests, or runtime evidence.

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ADR, C4 and requirements traceability · failure and recovery

FAILURE SIGNAL → LIKELY CAUSE

Diagram only	no decision rationale or consequences
Orphan test	evidence maps to no requirement
Stale contract	OpenAPI differs from deployed endpoint
Missing owner	component boundary cannot be governed

ACCEPTANCE EVIDENCE

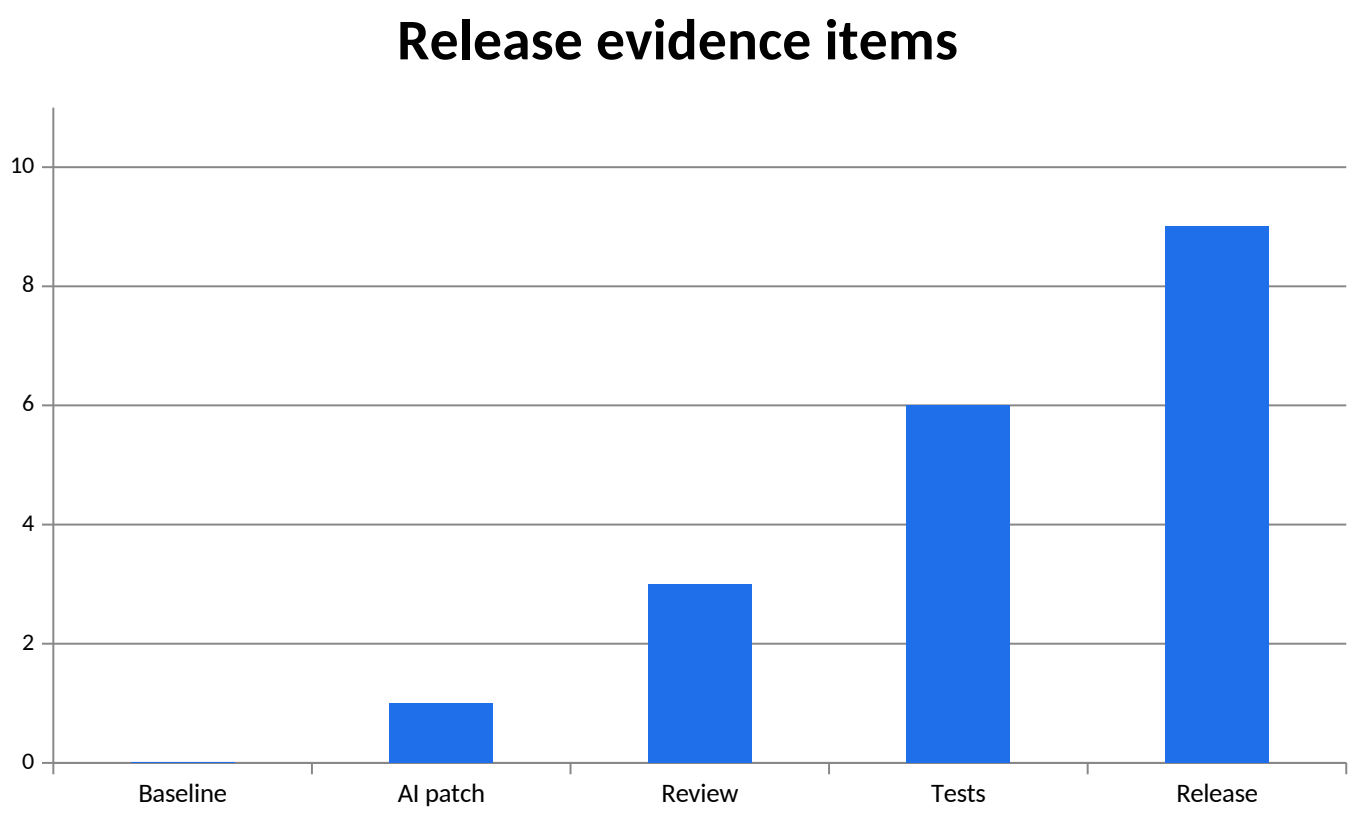
- ✓ Every LO requirement maps to component and test
- ✓ ADR names at least one rejected option
- ✓ OpenAPI hash matches released build

Record URL · command · status · build SHA

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TOPIC 5 · EDITABLE NATIVE CHART

LO5 evidence profile



WHAT THE DATA SHOWS

A deployable artifact is not a release until evidence closes the design record.

Worked-example values are illustrative evidence counts, not industry benchmarks. Edit the native chart with the class results.

EDITABLE CHART

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60 MIN

Activity 9: Prepare an Azure-Ready Release

Create a production-neutral publish artifact with validated configuration, health endpoints and smoke evidence.



YOU'LL PRODUCE **Release publish directory, health checks, deployment configuration matrix and smoke-test script**

Local snapshot: activities/activity-09-production-readiness/src/LocalCRM · full commands in the Learner Guide

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Activity 9 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Prepare LocalCRM for Azure App Service without performing a live deployment. Add validated typed options, /health/live and /health/ready, structured startup logging, and a bash smoke script. Do not add credentials, an Azure resource or a portal-only step.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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Activity 9 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: Release publish directory, health checks, deployment configuration matrix and smoke-test script.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

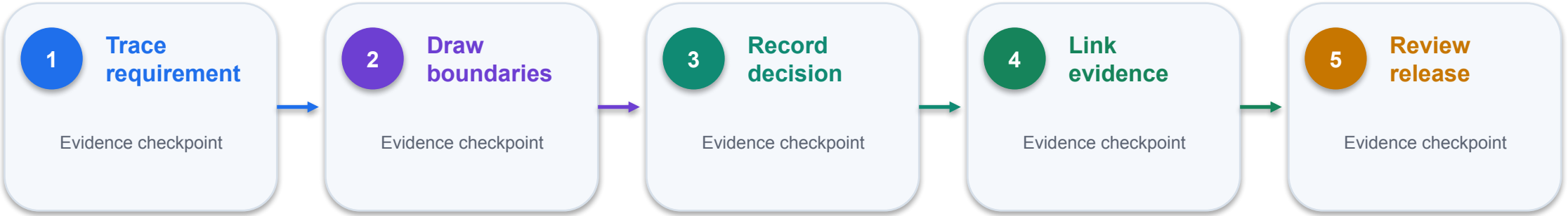
K5/A5 EVIDENCE

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55 MIN

Activity 10: Produce the Traceable Design and Release Record

Map user specifications to components, interface contracts, decisions and verification evidence.



YOU'LL PRODUCE **C4-style diagrams, ADR, requirements traceability matrix, API contract hash and release checklist**

Local snapshot: activities/activity-10-design-documentation/src/LocalCRM · full commands in the Learner Guide

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Activity 10 · reviewed AI contract

VIBE SCRIPT · COPY EXACTLY

Using requirements.md, component-map.md, openapi-v1.json and the Activity evidence, draft ADR-0001 and a traceability matrix. Every requirement must map to a component, interface or page, control, test/evidence and status. Mark gaps; do not invent evidence.

1	**Scope**	Only named files and component boundary
2	**Version**	net10.0 APIs and pinned packages
3	**Safety**	No credentials or hidden destructive step
4	**Evidence**	Build, test and HTTP acceptance named

The prompt is an input, not evidence. Evidence begins after the generated diff is inspected and the local snapshot passes its checks.

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TOPIC 5 · VERIFY

Activity 10 · acceptance and diagnosis

ACCEPTANCE

- ✓ The Activity's src/LocalCRM snapshot restores and starts locally.
- ✓ The declared output exists: C4-style diagrams, ADR, requirements traceability matrix, API contract hash and release checklist.
- ✓ Positive and negative paths produce the documented status or UI state.
- ✓ The AI diff contains no unrelated files, secrets, or unexplained dependency.

IF IT FAILS

1. Read the first build error
2. Confirm SDK/package version
3. Compare method + route + content type
4. Reset to the prior Activity snapshot

K5/A5 EVIDENCE

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LO5 · K5 / A5

Topic 5 · Evidence Recap

1 **Architecture**

Name the component boundary and owner.

2 **Mechanism**

Trace the runtime/data flow in order.

3 **Contract**

Point to fields, configuration, code or status semantics.

4 **Failure**

Recognise the named signal and likely cause.

5 **Verification**

Produce repeatable command or UI evidence.

6 **Activity link**

Use the cumulative LocalCRM snapshot for this LO.

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WHAT YOU CAN NOW PROVE

Course Synthesis

1

LO1

Determine basic software components using ASP.NET framework to meet functional specifications.

2

LO2

Utilize ASP.NET framework to develop web apps in adherence to organizational practices.

3

LO3

Select essential ASP.NET controls and features to meet software design requirements.

4

LO4

Examine the interoperability and functionality of ASP.NET software components.

5

LO5

Generate ASP.NET design documentation aligned with user specifications.

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DOWNLOAD · COMPLETE · SUBMIT

Courseware and Assessment on the LMS

1 **Courseware**

Download the current slides and Learner Guide.

2 **Activities**

Use the repository snapshots and mock CRM data.

3 **Assessment**

Complete WA then PP in the assigned window.

4 **Submission**

Upload answers and required practical evidence.

https://lms-tms.tertiaryinfotech.com/

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FINAL REMINDER · OPEN BOOK

Assessment

1

WA · 16:00–17:00

Five SAQ items assess K1–K5.

2

PP · 17:00–18:00

Five practical tasks assess A1–A5.

3

Attendance

Complete assessment digital attendance.

4

Submission

Upload evidence on the LMS and sign the record.

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FINAL MANDATORY SEQUENCE

Assessment Flow



Courseware and assessment: https://lms-tms.tertiaryinfotech.com/

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Digital Attendance (Mandatory)

1 **Scan**

Use the QR code shown by trainer/administrator.

2 **Submit**

Confirm the assessment attendance event.

3 **Check**

Resolve any missing record before leaving.

4 **Protect**

Do not photograph the assessment script.

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THANK YOU

Build it. Inspect it. Prove it.

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